CLINICAL STUDY REPORT

A Phase 3 Randomised, Placebo-controlled Study of Kelvaparib as Maintenance Therapy in Recurrent High-grade Serous Ovarian Carcinoma

Study KLV-7701 — Primary Analysis

Investigational product	Kelvaparib (KLV-3308), oral PARP1-selective inhibitor
Indication studied	High-grade serous ovarian carcinoma
Development phase	Phase 3
Sponsor	Marnholt Oncology GmbH
Protocol number	KLV-7701
First subject enrolled	7 July 2019
Data cut-off date	17 November 2021
Database lock	15 December 2021
Report date	1 June 2022
Report status	Primary Analysis
Good Clinical Practice	Conducted in accordance with the Declaration of Helsinki and ICH Good Clinical Practice.

2 SYNOPSIS

Name of sponsor	Marnholt Oncology GmbH
Name of finished product	KLV-3308
Name of active ingredient	Kelvaparib (KLV-3308)
Title of study	A Phase 3 Randomised, Placebo-controlled Study of Kelvaparib as Maintenance Therapy in Recurrent High-grade Serous Ovarian Carcinoma
Study number	KLV-7701
Phase of development	Phase 3
Studied period	7 July 2019 to 17 November 2021
Analysis	Primary Analysis
Objectives	Primary: progression-free survival by investigator assessment. Secondary: duration of response, overall survival and the incidence and severity of treatment-emergent adverse events, and pharmacokinetics.
Methodology	An open-label, multicentre, randomised study allocating subjects 2:1 to Kelvaparib 300 mg or Placebo. Randomisation is stratified by BRCA mutation status, platinum sensitivity and residual disease after surgery. Approximately 627 subjects.
Number of subjects	591 subjects were randomised and received at least one dose of study treatment (394 Kelvaparib 300 mg, 197 Placebo).
Diagnosis and main criteria for inclusion	Adults with histologically or cytologically confirmed recurrent ovarian carcinoma after prior platinum-based therapy, platinum-sensitive or platinum-resistant, measurable disease assessable by RECIST version 1.1, ECOG performance status 0 or 1, adequate organ function.
Test product, dose and mode of administration	Kelvaparib, film-coated tablets, administered orally twice daily on a continuous schedule, with weekly blood counts for the first month and monthly thereafter.
Duration of treatment	Until disease progression, unacceptable toxicity or withdrawal of consent. Median duration 114 days.
Criteria for evaluation - efficacy	Best overall response by RECIST version 1.1, and duration of response, progression-free survival, overall survival.
Criteria for evaluation - safety	Treatment-emergent adverse events graded by CTCAE version 5.0, laboratory parameters and vital signs.
Statistical methods	The primary endpoint is compared between treatment groups using a stratified log-rank test at a two-sided significance level of 0.05, with the hazard ratio and its 95 per cent confidence interval from a stratified Cox model. Key secondary endpoints are tested in a prespecified hierarchical order; significance is claimed for a secondary endpoint only if every preceding test in the hierarchy reached significance. Approximately 388 events provide 90 per cent power to detect a hazard ratio of 0.71.
Summary - efficacy	The overall response rate was Kelvaparib 300 mg 218 of 381 subjects (57.2 per cent, 95 per cent CI (52.1, 62.2)); Placebo 69 of 188 subjects (36.7 per cent, 95 per cent CI (29.8, 44.0)). Complete response was reported in 54 subjects (9.5 per cent) overall. Median progression-free survival 8.2 months.

Summary - safety	Grade 3 or higher events in 269 subjects (45.5 per cent), serious events in 138 (23.4 per cent). Nausea was the most frequent event (359 subjects, 60.7 per cent). Myelodysplastic syndrome or acute myeloid leukaemia, an event of special interest, occurred in 8 subjects (1.4 per cent). 84 discontinued because of an adverse event; 92 deaths were reported.
Conclusion	At this data cut-off, before the final analysis, progression-free survival favoured kelvaparib over placebo (stratified hazard ratio 0.72, 95 per cent CI 0.55 to 0.94; stratified log-rank p = 0.0140); the result is descriptive, because the statistical analysis plan spends no alpha before the final analysis. The overall response rate of 50.4 per cent is also descriptive.
Date of report	1 June 2022

3 TABLE OF CONTENTS

Section	Title	Page
1	Title page	1
2	Synopsis	2
3	Table of contents	4
4	List of abbreviations and definitions of terms	6
5	Ethics	6
5.1	Independent ethics committee or institutional review board	6
5.2	Ethical conduct of the study	7
5.3	Subject information and consent	7
6	Investigators and study administrative structure	7
7	Introduction	7
8	Study objectives	7
9	Investigational plan	8
9.1	Overall study design and plan	8
9.2	Discussion of study design, including choice of control	8
9.3	Selection of study population	8
9.4	Treatments	8
9.5	Efficacy and safety variables	8
9.6	Data quality assurance	9
9.7	Statistical methods planned in the protocol	9
9.8	Changes in the conduct of the study or the planned analyses	9
10	Study subjects	10
10.1	Disposition of subjects	10
10.1.1	Reasons for discontinuation	10
10.2	Protocol deviations	11
11	Efficacy evaluation	11
11.1	Data sets analysed	11
11.2	Demographic and other baseline characteristics	13
11.2.1	Discussion of each baseline characteristic	14
11.3	Measurements of treatment compliance	14
11.4	Efficacy results	16
11.4.1	Analysis of the primary endpoint	16
11.4.2	Analysis of time-to-event endpoints	17
11.4.3	Subgroup analyses	17
11.4.4	Efficacy conclusions	20
11.5	By-subject efficacy data	20
12	Safety evaluation	20
12.1	Extent of exposure	20

Section	Title	Page
12.2	Adverse events	20
12.2.1	Overview	20
12.2.1.1	Discussion of each event term	21
12.2.2	Events by system organ class and preferred term	27
12.2.3	Events of grade 3 or higher	30
12.2.4	Events considered related to study treatment	31
12.2.5	Timing of onset	32
12.2.6	Adverse events of special interest	32
12.3	Deaths, other serious adverse events and other significant events	34
12.3.1	Deaths	34
12.3.2	Serious adverse events	36
12.3.3	Events leading to discontinuation	37
12.3.4	Narratives	37
12.4	Clinical laboratory evaluation	37
12.4.1	Findings for each laboratory test	39
12.5	Vital signs, physical findings and other safety observations	40
12.5.1	Electrocardiograms	40
12.6	Safety conclusions	40
13	Discussion and overall conclusions	41
14	Tables and figures referred to but not in the text	42
14.1	Demographic data	43
14.2	Efficacy data	44
14.3	Safety data	53
14.3.10	Adverse events within subgroups	63
14.4	Pharmacokinetic data	70
15	Reference list	72
16	Appendices	73

4 LIST OF ABBREVIATIONS AND DEFINITIONS OF TERMS

Abbreviation	Definition
ALT	alanine aminotransferase
AST	aspartate aminotransferase
AUC0-12	area under the concentration-time curve over one dosing interval (0 to 12 hours)
AUC0-24	area under the concentration-time curve from 0 to 24 hours
BRCA	breast cancer susceptibility gene
BRCA1	breast cancer susceptibility gene 1
BRCA2	breast cancer susceptibility gene 2
CA-125	cancer antigen 125
CI	confidence interval
CL/F	apparent oral clearance
Cmax	maximum plasma concentration
CR	complete response
CTCAE	Common Terminology Criteria for Adverse Events
Ctrough	trough plasma concentration
CV	coefficient of variation
ECOG	Eastern Cooperative Oncology Group
ICH	International Council for Harmonisation
KLV-3308	kelvaparib (compound code)
MedDRA	Medical Dictionary for Regulatory Activities
NE	not estimable
PFS	progression-free survival
PR	partial response
Q1	first quartile
Q3	third quartile
RECIST	Response Evaluation Criteria in Solid Tumours
SD	stable disease
t1/2	terminal elimination half-life
TEAE	treatment-emergent adverse event
tmax	time to maximum plasma concentration

5 ETHICS

5.1 Independent ethics committee or institutional review board

The protocol, its amendments, the subject information sheet and the consent form were reviewed and approved by the responsible committee for every participating site before any subject was enrolled at that site. Approvals are held in the trial master file.

5.2 Ethical conduct of the study

The study was conducted in accordance with the protocol, the principles of the Declaration of Helsinki, the principles of Good Clinical Practice and the applicable regulatory requirements.

5.3 Subject information and consent

Written informed consent was obtained from every subject before any study-specific procedure. Where the information sheet was revised during the study, consent was obtained again from subjects still participating.

6 INVESTIGATORS AND STUDY ADMINISTRATIVE STRUCTURE

The study was conducted at the sites listed in Appendix 16.1.4, identified by number. Investigator names are not recorded in this report. The sponsor was responsible for the design of the study, for the statistical analysis and for the preparation of this report. Clinical monitoring and data management were performed by Steinbeck Clinical Research GmbH. Laboratory analyses were performed at a central laboratory. Response and progression were assessed by the investigator; no independent review was performed.

7 INTRODUCTION

This report describes study KLV-7701, A Phase 3 Randomised, Placebo-controlled Study of Kelvaparib as Maintenance Therapy in Recurrent High-grade Serous Ovarian Carcinoma. The study was conducted to evaluate Kelvaparib in subjects with recurrent ovarian carcinoma after prior platinum-based therapy, platinum-sensitive or platinum-resistant. The rationale for the study, for the dose and for the choice of endpoints is given in the protocol.

8 STUDY OBJECTIVES

Table 8-1 Study objectives

Type	Objective
Primary	Progression-free survival by investigator assessment
Safety	To characterise the safety and tolerability of Kelvaparib.

9 INVESTIGATIONAL PLAN

9.1 Overall study design and plan

An open-label, multicentre, randomised study allocating subjects 2:1 to Kelvaparib 300 mg or Placebo. Randomisation is stratified by BRCA mutation status, platinum sensitivity and residual disease after surgery. Approximately 627 subjects.

9.2 Discussion of study design, including choice of control

The design was appropriate to the question. A concurrent randomised control was used so that the treatment effect could be estimated without relying on an external comparison. The assessment interval was chosen so that the endpoint could be observed without imposing avoidable burden, and it was applied consistently across the study.

9.3 Selection of study population

Eligibility was defined in the protocol and is summarised in the synopsis (Section 2). Every subject was required to meet all inclusion criteria and none of the exclusion criteria, and no waivers were granted; 31 subjects enrolled after amendment 1 did not meet its performance status criterion (Section 10.2).

9.4 Treatments

Kelvaparib was administered orally twice daily on a continuous schedule. Dose modification followed the rules in the protocol. Study treatment was supplied by the sponsor and accountability was reconciled at each monitoring visit.

9.5 Efficacy and safety variables

Efficacy was assessed according to RECIST version 1.1. Safety was assessed by adverse event reporting, laboratory testing, vital signs and physical examination, with severity graded using the Common Terminology Criteria for Adverse Events and events coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 25.0; concomitant medications were coded using the WHO Drug Dictionary (WHODrug) Global B3, March 2022. Medical history, physical examination, electrocardiogram, coagulation, urinalysis and pregnancy test results were recorded at the sites but are not held in the study database and are not reported.

9.6 Data quality assurance

Data were recorded on an electronic case report form with programmed edit checks. Source data verification covered consent, eligibility, the primary endpoint and all serious adverse events for every subject. Coding was performed centrally and reviewed by the sponsor's safety physician. The database was locked after all queries were resolved and the analysis populations were agreed.

9.7 Statistical methods planned in the protocol

The primary endpoint is compared between treatment groups using a stratified log-rank test at a two-sided significance level of 0.05, with the hazard ratio and its 95 per cent confidence interval from a stratified Cox model. Key secondary endpoints are tested in a prespecified hierarchical order; significance is claimed for a secondary endpoint only if every preceding test in the hierarchy reached significance. Approximately 388 events provide 90 per cent power to detect a hazard ratio of 0.71.

9.8 Changes in the conduct of the study or the planned analyses

The protocol was first approved as version 1.0 on 23 June 2019. It was amended 3 times before the data cut-off of 17 November 2021:

Amendment 1 (protocol version 2.0, 10 July 2020): Revision of the eligibility criteria. Effect on the analysis, as stated in the amendment: the primary analysis is unchanged. No change to the analysis populations or to the primary endpoint is introduced by this amendment. Subjects enrolled under the previous criteria remain in the full analysis set.

Amendment 2 (protocol version 3.0, 30 January 2021): Revision following emerging safety data. Effect on the analysis, as stated in the amendment: the primary analysis is unchanged. No change to the analysis populations or to the primary endpoint is introduced by this amendment.

Amendment 3 (protocol version 4.0, 23 August 2021): Administrative and statistical update. Effect on the analysis, as stated in the amendment: the primary analysis is unchanged. The revised sample size is reflected in the statistical analysis plan, which is updated in parallel with this amendment.

The protocol of record for this report is version 4.0 (amendment 3, 23 August 2021). This report presents the analysis at a data cut-off of 17 November 2021, when 295 of the 388 planned progression-free survival events had occurred; the statistical analysis plan spends no alpha before the final analysis, so every result here is descriptive. The clinical study report of the final

analysis (clinical-study-report-klv-7701) is the report of record.

Changes to the planned analyses were documented in the statistical analysis plan before database lock. No analysis reported here was devised after the data were examined.

10 STUDY SUBJECTS

10.1 Disposition of subjects

591 subjects were enrolled and treated. Disposition is in Table 10-1. Of these, 385 discontinued study treatment, most commonly for withdrawal by subject.

Table 10-1 Subject Disposition – Safety Analysis Set

Disposition	Kelvaparib 300 mg (N=394)	Placebo (N=197)	Total (N=591)
Subjects treated, n	394	197	591
Treatment ongoing at data cut-off, n (%)	89 (22.6)	34 (17.3)	123 (20.8)
Discontinued study treatment, n (%)	317 (75.8)	67 (79.8)	384 (76.5)
Disease progression	30 (7.6)	35 (17.8)	65 (11.0)
Adverse event	25 (6.3)	11 (5.6)	36 (6.1)
Death	31 (7.9)	19 (9.6)	50 (8.5)
Withdrawal by subject	61 (15.5)	27 (13.7)	88 (14.9)
Physician decision	50 (12.7)	24 (12.2)	74 (12.5)
Lost to follow-up	30 (7.6)	8 (4.1)	38 (6.4)
Protocol deviation	24 (6.1)	10 (5.1)	34 (5.8)

Percentages are based on the number of subjects treated within each group.

10.1.1 Reasons for discontinuation

Withdrawal by subject. Recorded for 88 subjects (14.9 per cent of those treated). The date and the reason were recorded at the time the decision was taken, and each subject attended the end of treatment visit and the safety follow-up visit unless consent was withdrawn.

Physician decision. Recorded for 74 subjects (12.5 per cent of those treated). The date and the reason were recorded at the time the decision was taken, and each subject attended the end of treatment visit and the safety follow-up visit unless consent was withdrawn.

Disease progression. Recorded for 65 subjects (11.0 per cent of those treated). The date and the reason were recorded at the time the decision was taken, and each subject attended the end of treatment visit and the safety follow-up visit unless consent was withdrawn.

Death. Recorded for 50 subjects (8.5 per cent of those treated). The date and the reason were recorded at the time the decision was taken, and each subject attended the end of treatment visit and the safety follow-up visit unless consent was withdrawn.

Lost to follow-up. Recorded for 38 subjects (6.4 per cent of those treated). The date and the reason were recorded at the time the decision was taken, and each subject attended the end of treatment visit and the safety follow-up visit unless consent was withdrawn.

Protocol deviation. Recorded for 34 subjects (5.8 per cent of those treated). The date and the reason were recorded at the time the decision was taken, and each subject attended the end of treatment visit and the safety follow-up visit unless consent was withdrawn.

Adverse event. Recorded for 36 subjects (6.1 per cent of those treated). The date and the reason were recorded at the time the decision was taken, and each subject attended the end of treatment visit and the safety follow-up visit unless consent was withdrawn.

10.2 Protocol deviations

No protocol deviation dataset was collected in this study, and deviations were not classified as important or not important. The deviations held in the data are those recorded as the reason for discontinuing study treatment: 34 subjects (5.8 per cent of those treated, Table 10-1). These subjects are listed at Appendix 16.2.2. No monitoring record or deviation log is held in the study database, so deviations that did not lead to discontinuation of study treatment cannot be identified or listed. Amendment 1 (10 July 2020) limited eligibility to ECOG performance status 0 or 1; 31 of the 362 subjects who consented after it had a screening performance status of 2 and did not meet that criterion. They are eligibility deviations and remain in every analysis set. 21 of the same subjects had a baseline platelet count between 75 and 100 x 10^9/L; the screening value against which eligibility was judged is not held in the database.

11 EFFICACY EVALUATION

11.1 Data sets analysed

Table 11-1 Analysis sets

Analysis set	Subjects	Definition
Enrolled	591	All subjects assigned to treatment.
Safety	591	All subjects who received at least one dose.
Full analysis	569	All subjects in the safety analysis set with at least one post-baseline disease assessment.
Efficacy evaluable	515	Subjects with at least one evaluable post-baseline assessment.

11.2 Demographic and other baseline characteristics

Table 11-2 Demographic and Baseline Characteristics – Safety Analysis Set

Characteristic	Kelvaparib 300 mg (N=394)	Placebo (N=197)	Total (N=591)
Age (years)			
Mean (SD)	65.3 (9.01)	64.7 (9.44)	65.1 (9.15)
Median	66.0	65.0	65.0
Min, max	38, 88	42, 88	38, 88
>= 65 years, n (%)	215 (54.6)	104 (52.8)	319 (54.0)
Sex, n (%)			
Male	0 (0.0)	0 (0.0)	0 (0.0)
Female	394 (100.0)	197 (100.0)	591 (100.0)
Race, n (%)			
White	323 (82.0)	151 (76.6)	474 (80.2)
Asian	36 (9.1)	25 (12.7)	61 (10.3)
Black or African American	18 (4.6)	10 (5.1)	28 (4.7)
Other	17 (4.3)	11 (5.6)	28 (4.7)
ECOG performance status, n (%)			
0	181 (45.9)	98 (49.7)	279 (47.2)
1	176 (44.7)	83 (42.1)	259 (43.8)
2	37 (9.4)	16 (8.1)	53 (9.0)
BRCA mutation status, n (%)			
BRCA1 mutated	94 (23.9)	46 (23.4)	140 (23.7)
BRCA2 mutated	60 (15.2)	30 (15.2)	90 (15.2)
BRCA wild-type	240 (60.9)	121 (61.4)	361 (61.1)
Homologous recombination deficiency, n (%)	208 (52.8)	96 (48.7)	304 (51.4)
Platinum-sensitive disease, n (%)	252 (64.0)	129 (65.5)	381 (64.5)
Residual disease after cytoreductive surgery, n (%)	167 (42.4)	84 (42.6)	251 (42.5)
Number of prior lines of platinum-based therapy			
Median	2.0	2.0	2.0
>= 3, n (%)	119 (30.2)	58 (29.4)	177 (29.9)

SD = standard deviation; ECOG = Eastern Cooperative Oncology Group. Percentages are based on the number of subjects in the safety analysis set within each group.

11.2.1 Discussion of each baseline characteristic

Age. The median age was 65 years, with a range of 38 to 88. 319 subjects (54.0 per cent) were aged 65 or over. The age distribution is consistent with the population the eligibility criteria describe and no age-related restriction was applied beyond the requirement to be an adult.

Sex. All 591 subjects were female, as the eligibility criteria require.

Race. White: 474 (80.2 per cent); Asian: 61 (10.3 per cent); Black or African American: 28 (4.7 per cent); Other: 28 (4.7 per cent). Race was self-reported and was recorded in the categories used across the programme. The distribution reflects the territories in which the study recruited and should be borne in mind when generalising these results.

Performance status at baseline. score 0: 279 (47.2 per cent); score 1: 259 (43.8 per cent); score 2: 53 (9.0 per cent). Performance status is the single strongest predictor of tolerance in this population, and the distribution here is what the eligibility criteria were intended to produce.

Prior therapy. The median number of prior lines was 2, with a range of 1 to 4. Subjects entering with more prior therapy have by definition disease that has already been selected by exposure to more mechanisms, and the analyses by this characteristic are presented in the subgroup section.

BRCA mutation status. BRCA wild-type: 361 (61.1 per cent); BRCA1 mutated: 140 (23.7 per cent); BRCA2 mutated: 90 (15.2 per cent). Recorded at baseline and prespecified as a subgroup.

Homologous recombination status. Present in 304 subjects (51.4 per cent) at baseline. This factor was prespecified as a subgroup and is reported in the subgroup analyses.

Platinum sensitivity. Present in 381 subjects (64.5 per cent) at baseline. This factor was prespecified as a subgroup and is reported in the subgroup analyses.

Residual disease after surgery. Present in 251 subjects (42.5 per cent) at baseline. This factor was prespecified as a subgroup and is reported in the subgroup analyses.

11.3 Measurements of treatment compliance

The median duration of exposure was 114 days and the median relative dose intensity was 87.0 per cent. Exposure is summarised in Table 11-3.

Table 11-3 Extent of Exposure – Safety Analysis Set

Exposure parameter	Kelvaparib 300 mg (N=394)	Placebo (N=197)	Total (N=591)
Duration of treatment (days)			
Median	112.0	114.0	114.0
Q1, Q3	63.2, 159.0	55.0, 168.0	60.0, 163.5
Min, max	8, 340	3, 353	3, 353
Cycles started, median	4.0	4.0	4.0
Relative dose intensity (%), median	86.8	88.1	87.0
Any dose reduction, n (%)	69 (17.5)	37 (18.8)	106 (17.9)
Any dose interruption, n (%)	110 (27.9)	52 (26.4)	162 (27.4)

Q1 = first quartile; Q3 = third quartile. Relative dose intensity is the total dose received divided by the total dose planned over the treatment period.

All 591 subjects who were enrolled received at least one dose and are included in the safety population. The median duration of exposure was 114 days, with a range of 3 to 353.

The median relative dose intensity was 87.0 per cent. This is the measure that distinguishes a regimen that is genuinely tolerable from one that appears tolerable only because subjects are missing doses, and a value at this level indicates that most subjects received most of what was planned.

106 subjects (17.9 per cent) had at least one dose reduction and 162 (27.4 per cent) had at least one interruption. Reductions and interruptions followed the rules in the protocol; no subject was re-escalated after a reduction.

Exposure was sufficient to characterise both the efficacy and the safety of the regimen. A study in which most subjects stop early cannot support conclusions about either, and that is not the case here.

11.4 Efficacy results

11.4.1 Analysis of the primary endpoint

The primary endpoint was progression-free survival, defined as the time from the first dose of study treatment to the first documented disease progression by investigator assessment or death from any cause, whichever occurs first. 189 of 381 subjects in the Kelvaparib 300 mg group and 106 of 188 in the Placebo group had a progression-free survival event; median progression-free survival was 9.0 months with kelvaparib and 6.8 months with placebo (Table 11-5). At this data cut-off 295 of the 388 progression-free survival events required for the final analysis had occurred; the statistical analysis plan spends no alpha before the final analysis, so the stratified analysis (stratified hazard ratio 0.72, 95 per cent CI 0.55 to 0.94; stratified log-rank p = 0.0140) is descriptive, and the formal test of the primary endpoint is made at the final analysis. The overall response rate, a key secondary endpoint, was 287 of 569 subjects (50.4 per cent, 95 per cent CI (46.3, 54.6)). Because the testing hierarchy is prespecified, significance for a secondary endpoint is claimed only where every preceding test reached significance; p-values below that point are reported as nominal.

Table 11-4 Best Overall Response by RECIST 1.1 – Full Analysis Set

Best overall response	Kelvaparib 300 mg (N=381)	Placebo (N=188)	Total (N=569)
Complete response (CR)	40 (10.5)	14 (7.4)	54 (9.5)
Partial response (PR)	178 (46.7)	55 (29.3)	233 (40.9)
Stable disease	91 (23.9)	64 (34.0)	155 (27.2)
Progressive disease	36 (9.4)	37 (19.7)	73 (12.8)
Not evaluable	36 (9.4)	18 (9.6)	54 (9.5)
Objective response rate, n (%)	218 (57.2)	69 (36.7)	287 (50.4)
95% CI	(52.1, 62.2)	(29.8, 44.0)	(46.3, 54.6)

CI = confidence interval. Assessed according to RECIST version 1.1. Confidence intervals are exact (Clopper-Pearson).

Primary endpoint. The primary endpoint was progression-free survival, defined as the time from the first dose of study treatment to the first documented disease progression by investigator assessment or death from any cause, whichever occurs first. The overall response rate, a key secondary endpoint, was 287 of 569 subjects (50.4 per cent, 95 per cent CI (46.3, 54.6)). Because the testing hierarchy is prespecified, significance for a secondary endpoint is claimed only where every preceding test reached significance; p-values below that point are reported as nominal. The estimate and its confidence interval are in Table 11-4. The interval rather than the point estimate is what should be read, because it is the interval that states how precisely this

study has answered the question.

Response. 287 of 569 subjects (50.4 per cent, 95 per cent confidence interval (46.3, 54.6)) had a response. Responses were confirmed by a repeat assessment at least 4 weeks later; unconfirmed responses are recorded in the listings but are not counted here.

Duration of response. The median was 12.0 months. Follow-up was still accruing at the data cut-off, so this estimate is immature and will change as further events are observed.

Progression-free survival. The median was 8.2 months. Follow-up was still accruing at the data cut-off, so this estimate is immature and will change as further events are observed.

Overall survival. The median was 27.2 months. Follow-up was still accruing at the data cut-off, so this estimate is immature and will change as further events are observed.

Consistency. The subgroup tables give response and median progression-free survival for all subjects in each subgroup, both treatment groups together; they do not compare the treatment groups within a subgroup, so they do not show whether the effect was consistent across subgroups. None of those analyses was powered and none is adjusted for multiplicity.

11.4.2 Analysis of time-to-event endpoints

Table 11-5 Kaplan-Meier Estimates of Time-to-event Endpoints – Full Analysis Set

Endpoint	Kelvaparib 300 mg (N=381)	Placebo (N=188)	Total (N=569)
Duration of response, median (months)	12.0	9.3	12.0
25th, 75th percentile	(5.5, 21.8)	(2.9, 13.7)	(5.3, 15.9)
Progression-free survival, median (months)	9.0	6.8	8.2
25th, 75th percentile	(3.6, 14.8)	(2.8, 13.9)	(3.3, 14.5)
Overall survival, median (months)	27.2	NE	27.2
25th, 75th percentile	(22.5, NE)	NE	(22.5, NE)

NE = not estimable. Duration of response is analysed among responding subjects only. It is measured from the first dose to the end of response recorded in the analysis dataset; response and progression dates cannot be re-derived because no assessment-level tumour data were collected in the database.

11.4.3 Subgroup analyses

Response and median progression-free survival are shown below for all subjects in each prespecified subgroup, both treatment groups together. These analyses were not powered, and no conclusion is drawn from any single subgroup.

Table 11-6 Overall Response Rate and Progression-free Survival by BRCA Mutation Status – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
BRCA wild-type	347	177 (51.0)	(45.6, 56.4)	8.2
BRCA1 mutated	136	67 (49.3)	(40.6, 58.0)	7.6
BRCA2 mutated	86	43 (50.0)	(39.0, 61.0)	10.3

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 11-7 Overall Response Rate and Progression-free Survival by Homologous Recombination Status – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
Negative	277	132 (47.7)	(41.6, 53.7)	7.6
Positive	292	155 (53.1)	(47.2, 58.9)	8.7

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 11-8 Overall Response Rate and Progression-free Survival by Platinum Sensitivity – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
No	204	110 (53.9)	(46.8, 60.9)	8.4
Yes	365	177 (48.5)	(43.3, 53.8)	8.1

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 11-9 Overall Response Rate and Progression-free Survival by Residual Disease After Surgery – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
Absent	331	167 (50.5)	(44.9, 56.0)	8.5
Present	238	120 (50.4)	(43.9, 56.9)	7.5

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 11-10 Overall Response Rate and Progression-free Survival by Prior Lines of Platinum-based Therapy – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
1 to 2	397	196 (49.4)	(44.3, 54.4)	8.5
>= 3	172	91 (52.9)	(45.2, 60.5)	7.1

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 11-11 Overall Response Rate and Progression-free Survival by Age Category – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
< 65 years	260	143 (55.0)	(48.7, 61.2)	8.1
>= 65 years	309	144 (46.6)	(40.9, 52.3)	8.4

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 11-12 Overall Response Rate and Progression-free Survival by ECOG Performance Status – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
0	266	139 (52.3)	(46.1, 58.4)	7.6
1	252	122 (48.4)	(42.1, 54.8)	9.1
2	51	26 (51.0)	(36.6, 65.2)	8.4

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

BRCA Mutation Status. BRCA wild-type: 177 of 347 (51.0 per cent); BRCA1 mutated: 67 of 136 (49.3 per cent); BRCA2 mutated: 43 of 86 (50.0 per cent). The subgroups are of unequal size and none of these comparisons was powered, so the differences between them are presented to show consistency of direction rather than to support a conclusion about any one subgroup.

Homologous Recombination Status. Negative: 132 of 277 (47.7 per cent); Positive: 155 of 292 (53.1 per cent). The subgroups are of unequal size and none of these comparisons was powered, so the differences between them are presented to show consistency of direction rather than to support a conclusion about any one subgroup.

Platinum Sensitivity. No: 110 of 204 (53.9 per cent); Yes: 177 of 365 (48.5 per cent). The subgroups are of unequal size and none of these comparisons was powered, so the differences between them are presented to show consistency of direction rather than to support a conclusion about any one subgroup.

Residual Disease After Surgery. Absent: 167 of 331 (50.5 per cent); Present: 120 of 238 (50.4 per cent). The subgroups are of unequal size and none of these comparisons was powered, so the differences between them are presented to show consistency of direction rather than to support a conclusion about any one subgroup.

Prior Lines of Platinum-based Therapy. 1 to 2: 196 of 397 (49.4 per cent); >= 3: 91 of 172 (52.9 per cent). The subgroups are of unequal size and none of these comparisons was powered, so

the differences between them are presented to show consistency of direction rather than to support a conclusion about any one subgroup.

Age Category. < 65 years: 143 of 260 (55.0 per cent); >= 65 years: 144 of 309 (46.6 per cent). The subgroups are of unequal size and none of these comparisons was powered, so the differences between them are presented to show consistency of direction rather than to support a conclusion about any one subgroup.

ECOG Performance Status. 0: 139 of 266 (52.3 per cent); 1: 122 of 252 (48.4 per cent); 2: 26 of 51 (51.0 per cent). The subgroups are of unequal size and none of these comparisons was powered, so the differences between them are presented to show consistency of direction rather than to support a conclusion about any one subgroup.

11.4.4 Efficacy conclusions

At this data cut-off, before the final analysis, progression-free survival favoured kelvaparib over placebo (stratified hazard ratio 0.72, 95 per cent CI 0.55 to 0.94; stratified log-rank p = 0.0140); the result is descriptive, because the statistical analysis plan spends no alpha before the final analysis. The overall response rate of 50.4 per cent is also descriptive.

11.5 By-subject efficacy data

Individual response data for every subject are listed at Appendix 16.2.7, and individual time-to-event data at Appendix 16.2.8.

12 SAFETY EVALUATION

12.1 Extent of exposure

All 591 subjects received at least one dose. The extent of exposure is in Table 11-3.

12.2 Adverse events

12.2.1 Overview

Treatment-emergent adverse events were reported in the great majority of subjects. 269 subjects (45.5 per cent) had an event of grade 3 or higher, 138 (23.4 per cent) had a serious event and 84 (14.2 per cent) discontinued study treatment because of an event.

Table 12-1 Overview of Treatment-emergent Adverse Events – Safety Analysis Set

Category, n (%)	Kelvaparib 300 mg (N=394)	Placebo (N=197)	Total (N=591)
Any TEAE	391 (99.2)	197 (100.0)	588 (99.5)
Any treatment-related TEAE	383 (97.2)	190 (96.4)	573 (97.0)
Any grade >= 3 TEAE	172 (43.7)	97 (49.2)	269 (45.5)
Any serious TEAE	86 (21.8)	52 (26.4)	138 (23.4)
Any TEAE leading to dose reduction	69 (17.5)	36 (18.3)	105 (17.8)
Any TEAE leading to discontinuation	54 (13.7)	30 (15.2)	84 (14.2)
Any fatal TEAE	36 (9.1)	21 (10.7)	57 (9.6)

TEAE = treatment-emergent adverse event. A subject is counted once within each category.

12.2.1.1 Discussion of each event term

Each term reported in this study is discussed below with its incidence, the proportion reaching grade 3 or higher, the timing of first onset and the action taken.

Nausea was reported in 359 subjects (60.7 per cent), of whom 18 (3.0 per cent of the study population) had an event of grade 3 or higher, and 3 had an event recorded as serious. The median time to first onset was 31 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 1 subject. 62 subjects had an event that had not resolved at the data cut-off. The event falls within Gastrointestinal disorders.

Anaemia was reported in 284 subjects (48.1 per cent), of whom 90 (15.2 per cent of the study population) had an event of grade 3 or higher, and 52 had an event recorded as serious. The median time to first onset was 27 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 15 subjects. 5 events had a fatal outcome and are described in the narratives at Section 14.3.3. 47 subjects had an event that had not resolved at the data cut-off. The event falls within Blood and lymphatic system disorders.

Fatigue was reported in 275 subjects (46.5 per cent), of whom 15 (2.5 per cent of the study population) had an event of grade 3 or higher, and 4 had an event recorded as serious. The median time to first onset was 32 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 2 subjects. 53 subjects had an event that had not resolved at the data cut-off. The event falls within General disorders and administration site conditions.

Vomiting was reported in 150 subjects (25.4 per cent), of whom 8 (1.4 per cent of the study population) had an event of grade 3 or higher, and 1 had an event recorded as serious. The median time to first onset was 32 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 33 subjects had an event that had not resolved at the data cut-off. The event falls within Gastrointestinal disorders.

Constipation was reported in 149 subjects (25.2 per cent), of whom 2 (0.3 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 28 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 1 subject. 27 subjects had an event that had not resolved at the data cut-off. The event falls within Gastrointestinal disorders.

Decreased appetite was reported in 144 subjects (24.4 per cent), of whom 3 (0.5 per cent of the study population) had an event of grade 3 or higher, and 1 had an event recorded as serious. The median time to first onset was 36 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 1 subject. 25 subjects had an event that had not resolved at the data cut-off. The event falls within Metabolism and nutrition disorders.

Thrombocytopenia was reported in 144 subjects (24.4 per cent), of whom 32 (5.4 per cent of the study population) had an event of grade 3 or higher, and 13 had an event recorded as serious. The median time to first onset was 40 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 6 subjects. 5 events had a fatal outcome and are described in the narratives at Section 14.3.3. 22 subjects had an event that had not resolved at the data cut-off. The event falls within Blood and lymphatic system disorders.

Neutropenia was reported in 115 subjects (19.5 per cent), of whom 30 (5.1 per cent of the study population) had an event of grade 3 or higher, and 7 had an event recorded as serious. The median time to first onset was 35 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 7 subjects. 4 events had a fatal outcome and are described in the narratives at Section 14.3.3. 27 subjects had an event that had not resolved at the data cut-off. The event falls within Blood and lymphatic system disorders.

Headache was reported in 112 subjects (19.0 per cent), of whom 3 (0.5 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 24 days from

the first dose, so the event arose within the period of most intensive monitoring in the protocol. 20 subjects had an event that had not resolved at the data cut-off. The event falls within Nervous system disorders.

Diarrhoea was reported in 108 subjects (18.3 per cent), of whom 2 (0.3 per cent of the study population) had an event of grade 3 or higher, and 1 had an event recorded as serious. The median time to first onset was 38 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 21 subjects had an event that had not resolved at the data cut-off. The event falls within Gastrointestinal disorders.

Abdominal pain was reported in 107 subjects (18.1 per cent), of whom 3 (0.5 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 24 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 1 subject. 15 subjects had an event that had not resolved at the data cut-off. The event falls within Gastrointestinal disorders.

Dysgeusia was reported in 103 subjects (17.4 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 24 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 20 subjects had an event that had not resolved at the data cut-off. The event falls within Nervous system disorders.

Dizziness was reported in 93 subjects (15.7 per cent), of whom 2 (0.3 per cent of the study population) had an event of grade 3 or higher, and 1 had an event recorded as serious. The median time to first onset was 24 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 1 subject. 23 subjects had an event that had not resolved at the data cut-off. The event falls within Nervous system disorders.

Asthenia was reported in 81 subjects (13.7 per cent), of whom 4 (0.7 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 33 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 1 subject. 14 subjects had an event that had not resolved at the data cut-off. The event falls within General disorders and administration site conditions.

Cough was reported in 77 subjects (13.0 per cent), of whom 1 (0.2 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 33 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 16 subjects had an event that had not resolved at the data cut-off. The event falls within Respiratory, thoracic and mediastinal disorders.

Nasopharyngitis was reported in 72 subjects (12.2 per cent), of whom 1 (0.2 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 28 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 14 subjects had an event that had not resolved at the data cut-off. The event falls within Infections and infestations.

Back pain was reported in 71 subjects (12.0 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 24 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 11 subjects had an event that had not resolved at the data cut-off. The event falls within Musculoskeletal and connective tissue disorders.

Leukopenia was reported in 69 subjects (11.7 per cent), of whom 8 (1.4 per cent of the study population) had an event of grade 3 or higher, and 2 had an event recorded as serious. The median time to first onset was 29 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 3 subjects. 15 subjects had an event that had not resolved at the data cut-off. The event falls within Blood and lymphatic system disorders.

Insomnia was reported in 68 subjects (11.5 per cent), of whom 1 (0.2 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 38 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 11 subjects had an event that had not resolved at the data cut-off. The event falls within Psychiatric disorders.

Arthralgia was reported in 65 subjects (11.0 per cent), of whom 1 (0.2 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 34 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 10 subjects had an event that had not resolved at the data cut-off. The event falls within Musculoskeletal and connective tissue disorders.

Dyspepsia was reported in 58 subjects (9.8 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 29 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 10 subjects had an event that had not resolved at the data cut-off. The event falls within Gastrointestinal disorders.

Urinary tract infection was reported in 58 subjects (9.8 per cent), of whom 4 (0.7 per cent of the study population) had an event of grade 3 or higher, and 1 had an event recorded as serious. The median time to first onset was 30 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 11 subjects had an event that had not resolved at the data cut-off. The event falls within Infections and infestations.

Rash was reported in 55 subjects (9.3 per cent), of whom 2 (0.3 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 33 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 11 subjects had an event that had not resolved at the data cut-off. The event falls within Skin and subcutaneous tissue disorders.

Alanine aminotransferase increased was reported in 52 subjects (8.8 per cent), of whom 6 (1.0 per cent of the study population) had an event of grade 3 or higher, and 2 had an event recorded as serious. The median time to first onset was 30 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 14 subjects had an event that had not resolved at the data cut-off. The event falls within Investigations.

Dyspnoea was reported in 49 subjects (8.3 per cent), of whom 6 (1.0 per cent of the study population) had an event of grade 3 or higher, and 2 had an event recorded as serious. The median time to first onset was 32 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 3 subjects. 2 events had a fatal outcome and are described in the narratives at Section 14.3.3. 7 subjects had an event that had not resolved at the data cut-off. The event falls within Respiratory, thoracic and mediastinal disorders.

Oedema peripheral was reported in 45 subjects (7.6 per cent), of whom 1 (0.2 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 7 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 6 subjects had an event that had not resolved at the data cut-off. The event falls within General disorders and administration site conditions.

Lymphopenia was reported in 42 subjects (7.1 per cent), of whom 8 (1.4 per cent of the study population) had an event of grade 3 or higher, and 1 had an event recorded as serious. The median time to first onset was 46 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 2 subjects. 2 subjects had an event that had not resolved at the data cut-off. The event falls within Blood and lymphatic system disorders.

Myalgia was reported in 41 subjects (6.9 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 24 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 13 subjects had an event that had not resolved at the data cut-off. The event falls within Musculoskeletal and connective tissue disorders.

12.2.2 Events by system organ class and preferred term

Table 12-2 Treatment-emergent Adverse Events Occurring in at Least 10% of Subjects by System Organ Class and Preferred Term – Safety Analysis Set

System organ class / Preferred term	Kelvaparib 300 mg (N=394)	Placebo (N=197)	Total (N=591)
Gastrointestinal disorders	335 (85.0)	166 (84.3)	501 (84.8)
Nausea	236 (59.9)	123 (62.4)	359 (60.7)
Vomiting	103 (26.1)	47 (23.9)	150 (25.4)
Constipation	94 (23.9)	55 (27.9)	149 (25.2)
Diarrhoea	74 (18.8)	34 (17.3)	108 (18.3)
Abdominal pain	75 (19.0)	32 (16.2)	107 (18.1)
Blood and lymphatic system disorders	288 (73.1)	145 (73.6)	433 (73.3)
Anaemia	185 (47.0)	99 (50.3)	284 (48.1)
Thrombocytopenia	97 (24.6)	47 (23.9)	144 (24.4)
Neutropenia	75 (19.0)	40 (20.3)	115 (19.5)
Leukopenia	43 (10.9)	26 (13.2)	69 (11.7)
General disorders and administration site conditions	236 (59.9)	118 (59.9)	354 (59.9)
Fatigue	185 (47.0)	90 (45.7)	275 (46.5)
Asthenia	50 (12.7)	31 (15.7)	81 (13.7)
Nervous system disorders	169 (42.9)	84 (42.6)	253 (42.8)
Headache	74 (18.8)	38 (19.3)	112 (19.0)
Dysgeusia	61 (15.5)	42 (21.3)	103 (17.4)
Dizziness	67 (17.0)	26 (13.2)	93 (15.7)
Musculoskeletal and connective tissue disorders	107 (27.2)	61 (31.0)	168 (28.4)
Back pain	45 (11.4)	26 (13.2)	71 (12.0)
Arthralgia	43 (10.9)	22 (11.2)	65 (11.0)
Infections and infestations	108 (27.4)	58 (29.4)	166 (28.1)
Nasopharyngitis	51 (12.9)	21 (10.7)	72 (12.2)
Investigations	100 (25.4)	53 (26.9)	153 (25.9)
Metabolism and nutrition disorders	90 (22.8)	54 (27.4)	144 (24.4)
Decreased appetite	90 (22.8)	54 (27.4)	144 (24.4)
Respiratory, thoracic and mediastinal disorders	91 (23.1)	46 (23.4)	137 (23.2)
Cough	52 (13.2)	25 (12.7)	77 (13.0)
Psychiatric disorders	41 (10.4)	27 (13.7)	68 (11.5)
Insomnia	41 (10.4)	27 (13.7)	68 (11.5)

Subjects are counted once per preferred term and once per system organ class at their highest reported grade. MedDRA version 25.0.

Gastrointestinal disorders. Events in this class were reported in 501 subjects (84.8 per cent), and in 39 (6.6 per cent) at grade 3 or higher. The terms reported most often were nausea, vomiting, constipation. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Blood and lymphatic system disorders. Events in this class were reported in 433 subjects (73.3 per cent), and in 156 (26.4 per cent) at grade 3 or higher. The terms reported most often were anaemia, thrombocytopenia, neutropenia. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

General disorders and administration site conditions. Events in this class were reported in 354 subjects (59.9 per cent), and in 22 (3.7 per cent) at grade 3 or higher. The terms reported most often were fatigue, asthenia, oedema peripheral. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Nervous system disorders. Events in this class were reported in 253 subjects (42.8 per cent), and in 5 (0.8 per cent) at grade 3 or higher. The terms reported most often were headache, dysgeusia, dizziness. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Musculoskeletal and connective tissue disorders. Events in this class were reported in 168 subjects (28.4 per cent), and in 1 (0.2 per cent) at grade 3 or higher. The terms reported most often were back pain, arthralgia, myalgia. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Infections and infestations. Events in this class were reported in 166 subjects (28.1 per cent), and in 38 (6.4 per cent) at grade 3 or higher. The terms reported most often were nasopharyngitis, urinary tract infection, pneumonia. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Investigations. Events in this class were reported in 153 subjects (25.9 per cent), and in 12 (2.0 per cent) at grade 3 or higher. The terms reported most often were alanine aminotransferase increased, aspartate aminotransferase increased, blood creatinine increased. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Metabolism and nutrition disorders. Events in this class were reported in 144 subjects (24.4 per cent), and in 3 (0.5 per cent) at grade 3 or higher. The terms reported most often were decreased appetite. The pattern is consistent with what is anticipated for oral PARP1-selective

inhibitor in this population.

Respiratory, thoracic and mediastinal disorders. Events in this class were reported in 137 subjects (23.2 per cent), and in 16 (2.7 per cent) at grade 3 or higher. The terms reported most often were cough, dyspnoea, pneumonitis. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Psychiatric disorders. Events in this class were reported in 68 subjects (11.5 per cent), and in 1 (0.2 per cent) at grade 3 or higher. The terms reported most often were insomnia. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Skin and subcutaneous tissue disorders. Events in this class were reported in 55 subjects (9.3 per cent), and in 2 (0.3 per cent) at grade 3 or higher. The terms reported most often were rash. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Vascular disorders. Events in this class were reported in 34 subjects (5.8 per cent), and in 3 (0.5 per cent) at grade 3 or higher. The terms reported most often were hypertension. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Neoplasms benign, malignant and unspecified. Events in this class were reported in 18 subjects (3.0 per cent), and in 16 (2.7 per cent) at grade 3 or higher. The terms reported most often were malignant neoplasm progression, myelodysplastic syndrome. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

12.2.3 Events of grade 3 or higher

Table 12-3 Treatment-emergent Adverse Events of Grade 3 or Higher by System Organ Class and Preferred Term – Safety Analysis Set

System organ class / Preferred term	Kelvaparib 300 mg (N=394)	Placebo (N=197)	Total (N=591)
Blood and lymphatic system disorders	100 (25.4)	56 (28.4)	156 (26.4)
Anaemia	53 (13.5)	37 (18.8)	90 (15.2)
Gastrointestinal disorders	25 (6.3)	14 (7.1)	39 (6.6)
Infections and infestations	24 (6.1)	14 (7.1)	38 (6.4)
General disorders and administration site conditions	14 (3.6)	8 (4.1)	22 (3.7)
Neoplasms benign, malignant and unspecified	8 (2.0)	8 (4.1)	16 (2.7)
Respiratory, thoracic and mediastinal disorders	8 (2.0)	8 (4.1)	16 (2.7)
Investigations	8 (2.0)	4 (2.0)	12 (2.0)
Nervous system disorders	2 (0.5)	3 (1.5)	5 (0.8)
Vascular disorders	3 (0.8)	0 (0.0)	3 (0.5)
Metabolism and nutrition disorders	1 (0.3)	2 (1.0)	3 (0.5)
Skin and subcutaneous tissue disorders	2 (0.5)	0 (0.0)	2 (0.3)
Musculoskeletal and connective tissue disorders	1 (0.3)	0 (0.0)	1 (0.2)
Psychiatric disorders	1 (0.3)	0 (0.0)	1 (0.2)

Subjects are counted once per preferred term and once per system organ class at their highest reported grade. MedDRA version 25.0.

12.2.4 Events considered related to study treatment

Table 12-4 Treatment-related Treatment-emergent Adverse Events Occurring in at Least 5% of Subjects – Safety Analysis Set

Preferred term	Kelvaparib 300 mg (N=394)	Placebo (N=197)	Total (N=591)
Nausea	165 (41.9)	77 (39.1)	242 (40.9)
Anaemia	105 (26.6)	64 (32.5)	169 (28.6)
Fatigue	99 (25.1)	46 (23.4)	145 (24.5)
Vomiting	67 (17.0)	31 (15.7)	98 (16.6)
Constipation	63 (16.0)	31 (15.7)	94 (15.9)
Decreased appetite	60 (15.2)	31 (15.7)	91 (15.4)
Thrombocytopenia	54 (13.7)	28 (14.2)	82 (13.9)
Headache	50 (12.7)	28 (14.2)	78 (13.2)
Neutropenia	47 (11.9)	28 (14.2)	75 (12.7)
Diarrhoea	45 (11.4)	26 (13.2)	71 (12.0)
Abdominal pain	42 (10.7)	23 (11.7)	65 (11.0)
Dysgeusia	38 (9.6)	23 (11.7)	61 (10.3)
Dizziness	39 (9.9)	20 (10.2)	59 (10.0)
Asthenia	34 (8.6)	18 (9.1)	52 (8.8)
Cough	34 (8.6)	17 (8.6)	51 (8.6)
Nasopharyngitis	33 (8.4)	14 (7.1)	47 (8.0)
Back pain	28 (7.1)	18 (9.1)	46 (7.8)
Insomnia	27 (6.9)	14 (7.1)	41 (6.9)
Leukopenia	25 (6.3)	16 (8.1)	41 (6.9)
Urinary tract infection	26 (6.6)	13 (6.6)	39 (6.6)
Rash	18 (4.6)	20 (10.2)	38 (6.4)
Dyspepsia	24 (6.1)	13 (6.6)	37 (6.3)
Arthralgia	21 (5.3)	16 (8.1)	37 (6.3)
Alanine aminotransferase increased	19 (4.8)	12 (6.1)	31 (5.2)

Relationship assessed by the investigator. Percentages are based on the number of subjects treated in each group. Overall N = 591.

12.2.5 Timing of onset

Table 12-5 Time to Onset and Resolution of the Most Frequent Treatment-emergent Adverse Events – Safety Analysis Set

Preferred term	Events, n	Median time to onset (days)	Resolved, n (%)	Median time to resolution (days)
Nausea	359	31.0	297 (82.7)	11.0
Anaemia	285	27.0	238 (83.5)	10.0
Fatigue	275	32.0	222 (80.7)	10.0
Vomiting	150	32.0	117 (78.0)	11.0
Constipation	149	28.0	122 (81.9)	9.0
Thrombocytopenia	145	40.0	123 (84.8)	12.0
Decreased appetite	144	36.0	119 (82.6)	10.0
Neutropenia	116	35.0	89 (76.7)	10.0
Headache	112	23.5	92 (82.1)	10.0
Diarrhoea	108	37.5	87 (80.6)	11.0
Abdominal pain	107	24.0	92 (86.0)	12.0
Dysgeusia	103	24.0	83 (80.6)	10.0
Dizziness	93	24.0	70 (75.3)	12.5
Asthenia	81	33.0	67 (82.7)	11.0
Cough	77	33.0	61 (79.2)	10.0
Nasopharyngitis	72	27.5	58 (80.6)	11.5
Back pain	71	24.0	60 (84.5)	9.5
Leukopenia	69	29.0	54 (78.3)	11.0
Insomnia	68	37.5	57 (83.8)	8.0
Arthralgia	65	34.0	55 (84.6)	8.0

Events are counted at the event level rather than the subject level. NE = not estimable.

12.2.6 Adverse events of special interest

Three categories of event were designated of special interest for this programme on the basis of the mechanism of action and the findings of the first-in-human study. Each is a grouping of preferred terms rather than a single term, and a subject contributing more than one constituent term is counted once.

Table 12-6 Adverse Events of Special Interest and Their Component Terms - Safety Analysis Set

Event of special interest	Constituent preferred terms	Minimum grade	Subjects, n (%)	Grade >= 3, n (%)	Serious, n (%)
Myelodysplastic syndrome or acute myeloid leukaemia	Myelodysplastic syndrome	Any	8 (1.4)	6 (1.0)	5 (0.8)
Pneumonitis	Pneumonitis	Any	20 (3.4)	6 (1.0)	6 (1.0)
Severe anaemia	Anaemia	3	90 (15.2)	90 (15.2)	52 (8.8)

A subject reporting more than one constituent term is counted once within the category. Minimum grade is the lowest CTCAE grade at which a constituent term qualifies.

Myelodysplastic syndrome or acute myeloid leukaemia

Myelodysplastic syndrome or acute myeloid leukaemia is derived from the preferred term Myelodysplastic syndrome. The category was reported in 8 of 591 subjects (1.4 per cent), of whom 6 (1.0 per cent) had an event of grade 3 or higher and 3 had a fatal outcome.

The median time to first onset was 29 days from the first dose. Events were managed by the measures set out in the protocol; no subject was withdrawn from the study for this reason without the event first being reported to the sponsor within 24 hours.

Pneumonitis

Pneumonitis is reported as the coded preferred term. The category was reported in 20 of 591 subjects (3.4 per cent), of whom 6 (1.0 per cent) had an event of grade 3 or higher and 2 had a fatal outcome.

The median time to first onset was 39 days from the first dose. Events were managed by the measures set out in the protocol; no subject was withdrawn from the study for this reason without the event first being reported to the sponsor within 24 hours.

Severe anaemia

Severe anaemia is derived from the preferred term Anaemia at grade 3 or above. The category was reported in 90 of 591 subjects (15.2 per cent), of whom 90 (15.2 per cent) had an event of grade 3 or higher and 5 had a fatal outcome.

The median time to first onset was 34 days from the first dose. Events were managed by the measures set out in the protocol; no subject was withdrawn from the study for this reason without the event first being reported to the sponsor within 24 hours.

Nausea was the most frequent event overall, reported in 359 subjects (60.7 per cent), of whom 18 (3.0 per cent) had an event of grade 3 or higher.

12.3 Deaths, other serious adverse events and other significant events

12.3.1 Deaths

92 subjects died. The causes recorded are in Table 12-7. Deaths attributed to an adverse event are described in the narratives at Section 14.3.3.

Table 12-7 Deaths – Safety Analysis Set

Category, n (%)	Kelvaparib 300 mg (N=394)	Placebo (N=197)	Total (N=591)
Any death	61 (15.5)	31 (15.7)	92 (15.6)
Death within 30 days of last dose	31 (7.9)	19 (9.6)	50 (8.5)
Primary cause of death			
Disease progression	23 (5.8)	8 (4.1)	31 (5.2)
Adverse event	36 (9.1)	21 (10.7)	57 (9.6)
Other	2 (0.5)	2 (1.0)	4 (0.7)

Percentages are based on the number of subjects treated within each group.

92 of 591 subjects (15.6 per cent) died. The primary cause recorded for each is below. Every death has a dated record in the disposition dataset and, where an adverse event was the cause, a fatal event in the adverse event dataset.

Disease progression. Recorded as the primary cause for 31 subjects (33.7 per cent of those who died). These deaths were not attributed to an adverse event and are recorded as disease outcomes.

Adverse event. Recorded as the primary cause for 57 subjects (62.0 per cent of those who died). Each of these subjects has a corresponding fatal adverse event and a narrative at Section 14.3.3.

Other. Recorded as the primary cause for 4 subjects (4.3 per cent of those who died). These deaths were not attributed to an adverse event and are recorded as disease outcomes.

12.3.2 Serious adverse events

Table 12-8 Serious Treatment-emergent Adverse Events by Preferred Term – Safety Analysis Set

Preferred term	Kelvaparib 300 mg (N=394)	Placebo (N=197)	Total (N=591)
Anaemia	31 (7.9)	21 (10.7)	52 (8.8)
Pneumonia	11 (2.8)	3 (1.5)	14 (2.4)
Thrombocytopenia	12 (3.0)	1 (0.5)	13 (2.2)
Malignant neoplasm progression	5 (1.3)	5 (2.5)	10 (1.7)
Sepsis	5 (1.3)	4 (2.0)	9 (1.5)
Neutropenia	4 (1.0)	3 (1.5)	7 (1.2)
Pneumonitis	3 (0.8)	3 (1.5)	6 (1.0)
Myelodysplastic syndrome	3 (0.8)	2 (1.0)	5 (0.8)
Fatigue	3 (0.8)	1 (0.5)	4 (0.7)
Intestinal obstruction	2 (0.5)	2 (1.0)	4 (0.7)
Ascites	2 (0.5)	1 (0.5)	3 (0.5)
Nausea	2 (0.5)	1 (0.5)	3 (0.5)
Alanine aminotransferase increased	1 (0.3)	1 (0.5)	2 (0.3)
Dyspnoea	2 (0.5)	0 (0.0)	2 (0.3)
Leukopenia	0 (0.0)	2 (1.0)	2 (0.3)
Pulmonary embolism	1 (0.3)	1 (0.5)	2 (0.3)
Aspartate aminotransferase increased	1 (0.3)	0 (0.0)	1 (0.2)
Decreased appetite	0 (0.0)	1 (0.5)	1 (0.2)
Diarrhoea	1 (0.3)	0 (0.0)	1 (0.2)
Dizziness	0 (0.0)	1 (0.5)	1 (0.2)
Hypertension	1 (0.3)	0 (0.0)	1 (0.2)
Lymphopenia	1 (0.3)	0 (0.0)	1 (0.2)
Pleural effusion	0 (0.0)	1 (0.5)	1 (0.2)
Urinary tract infection	1 (0.3)	0 (0.0)	1 (0.2)
Vomiting	1 (0.3)	0 (0.0)	1 (0.2)

Subjects are counted once per preferred term. Percentages are based on N = 591.

12.3.3 Events leading to discontinuation

Table 12-9 Treatment-emergent Adverse Events Leading to Discontinuation of Study Treatment – Safety Analysis Set

Preferred term	Kelvaparib 300 mg (N=394)	Placebo (N=197)	Total (N=591)
Anaemia	9 (2.3)	6 (3.0)	15 (2.5)
Malignant neoplasm progression	5 (1.3)	5 (2.5)	10 (1.7)
Sepsis	6 (1.5)	3 (1.5)	9 (1.5)
Pneumonia	7 (1.8)	1 (0.5)	8 (1.4)
Neutropenia	5 (1.3)	2 (1.0)	7 (1.2)
Thrombocytopenia	6 (1.5)	0 (0.0)	6 (1.0)
Intestinal obstruction	2 (0.5)	2 (1.0)	4 (0.7)
Ascites	2 (0.5)	1 (0.5)	3 (0.5)
Dyspnoea	3 (0.8)	0 (0.0)	3 (0.5)
Leukopenia	1 (0.3)	2 (1.0)	3 (0.5)
Myelodysplastic syndrome	1 (0.3)	2 (1.0)	3 (0.5)
Fatigue	2 (0.5)	0 (0.0)	2 (0.3)
Lymphopenia	2 (0.5)	0 (0.0)	2 (0.3)
Pneumonitis	0 (0.0)	2 (1.0)	2 (0.3)
Pulmonary embolism	1 (0.3)	1 (0.5)	2 (0.3)
Abdominal pain	1 (0.3)	0 (0.0)	1 (0.2)
Asthenia	1 (0.3)	0 (0.0)	1 (0.2)
Constipation	0 (0.0)	1 (0.5)	1 (0.2)
Decreased appetite	1 (0.3)	0 (0.0)	1 (0.2)
Dizziness	0 (0.0)	1 (0.5)	1 (0.2)
Nausea	1 (0.3)	0 (0.0)	1 (0.2)
Pleural effusion	0 (0.0)	1 (0.5)	1 (0.2)
Pyrexia	1 (0.3)	0 (0.0)	1 (0.2)

Percentages are based on N = 591.

12.3.4 Narratives

Narratives for every death, every serious adverse event and every withdrawal due to an adverse event are provided at Section 14.3.3, bound separately.

12.4 Clinical laboratory evaluation

Laboratory results were graded against the central laboratory reference ranges. Shifts from baseline to the worst post-baseline grade are in Table 12-10. Individual laboratory data are listed at Appendix 16.2.10.

Table 12-10 Shift from Baseline to Worst Post-baseline CTCAE Grade for Selected Laboratory Parameters – Safety Analysis Set

Laboratory parameter	Kelvaparib 300 mg: worst grade 3 n (%)	Kelvaparib 300 mg: worst grade 4 n (%)	Kelvaparib 300 mg: shift >= 2 grades n (%)	Placebo: worst grade 3 n (%)	Placebo: worst grade 4 n (%)	Placebo: shift >= 2 grades n (%)	Total: worst grade 3 n (%)	Total: worst grade 4 n (%)	Total: shift >= 2 grades n (%)
Haemoglobin	52 (13.2)	12 (3.0)	112 (28.4)	33 (16.8)	9 (4.6)	66 (33.5)	85 (14.4)	21 (3.6)	178 (30.1)
Platelets	22 (5.6)	5 (1.3)	62 (15.7)	15 (7.6)	2 (1.0)	28 (14.2)	37 (6.3)	7 (1.2)	90 (15.2)
Neutrophils	30 (7.6)	14 (3.6)	70 (17.8)	16 (8.1)	10 (5.1)	38 (19.3)	46 (7.8)	24 (4.1)	108 (18.3)
Lymphocytes	28 (7.1)	9 (2.3)	62 (15.7)	8 (4.1)	2 (1.0)	27 (13.7)	36 (6.1)	11 (1.9)	89 (15.1)
CA-125	41 (10.4)	10 (2.5)	112 (28.4)	26 (13.2)	7 (3.6)	52 (26.4)	67 (11.3)	17 (2.9)	164 (27.7)
ALT	16 (4.1)	2 (0.5)	39 (9.9)	9 (4.6)	1 (0.5)	17 (8.6)	25 (4.2)	3 (0.5)	56 (9.5)
AST	13 (3.3)	1 (0.3)	25 (6.3)	8 (4.1)	3 (1.5)	17 (8.6)	21 (3.6)	4 (0.7)	42 (7.1)
Creatinine	20 (5.1)	6 (1.5)	40 (10.2)	8 (4.1)	1 (0.5)	17 (8.6)	28 (4.7)	7 (1.2)	57 (9.6)
Lactate dehydrogenase	22 (5.6)	5 (1.3)	51 (12.9)	10 (5.1)	4 (2.0)	20 (10.2)	32 (5.4)	9 (1.5)	71 (12.0)
Albumin	10 (2.5)	4 (1.0)	23 (5.8)	4 (2.0)	3 (1.5)	18 (9.1)	14 (2.4)	7 (1.2)	41 (6.9)

CTCAE = Common Terminology Criteria for Adverse Events, version 5.0. Percentages are based on the number of subjects treated in each group. Overall N = 591.

12.4.1 Findings for each laboratory test

ALT. An abnormality of any grade was present at baseline in 168 subjects (28.4 per cent) and post-baseline in 221 (37.4 per cent). 112 subjects (19.0 per cent) shifted to a worse grade than their own baseline, and 28 (4.7 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

AST. An abnormality of any grade was present at baseline in 150 subjects (25.4 per cent) and post-baseline in 209 (35.4 per cent). 97 subjects (16.4 per cent) shifted to a worse grade than their own baseline, and 25 (4.2 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

Albumin. An abnormality of any grade was present at baseline in 160 subjects (27.1 per cent) and post-baseline in 199 (33.7 per cent). 88 subjects (14.9 per cent) shifted to a worse grade than their own baseline, and 21 (3.6 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

CA-125. An abnormality of any grade was present at baseline in 158 subjects (26.7 per cent) and post-baseline in 392 (66.3 per cent). 354 subjects (59.9 per cent) shifted to a worse grade than their own baseline, and 84 (14.2 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

Creatinine. An abnormality of any grade was present at baseline in 194 subjects (32.8 per cent) and post-baseline in 255 (43.1 per cent). 123 subjects (20.8 per cent) shifted to a worse grade than their own baseline, and 35 (5.9 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

Haemoglobin. An abnormality of any grade was present at baseline in 180 subjects (30.5 per cent) and post-baseline in 419 (70.9 per cent). 374 subjects (63.3 per cent) shifted to a worse grade than their own baseline, and 106 (17.9 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

Lactate dehydrogenase. An abnormality of any grade was present at baseline in 172 subjects (29.1 per cent) and post-baseline in 260 (44.0 per cent). 158 subjects (26.7 per cent) shifted to a worse grade than their own baseline, and 41 (6.9 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

Lymphocytes. An abnormality of any grade was present at baseline in 176 subjects (29.8 per cent) and post-baseline in 285 (48.2 per cent). 187 subjects (31.6 per cent) shifted to a worse grade than their own baseline, and 47 (8.0 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

Neutrophils. An abnormality of any grade was present at baseline in 189 subjects (32.0 per cent) and post-baseline in 321 (54.3 per cent). 229 subjects (38.7 per cent) shifted to a worse grade than their own baseline, and 70 (11.8 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

Platelets. An abnormality of any grade was present at baseline in 152 subjects (25.7 per cent) and post-baseline in 288 (48.7 per cent). 214 subjects (36.2 per cent) shifted to a worse grade than their own baseline, and 44 (7.4 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

12.5 Vital signs, physical findings and other safety observations

No clinically significant pattern of change in vital signs was observed. Individual vital sign data are listed at Appendix 16.2.11.

12.5.1 Electrocardiograms

The protocol specified electrocardiograms, but no electrocardiogram data are held in the study database, so no electrocardiogram results are reported.

12.6 Safety conclusions

The safety profile observed was consistent with that anticipated for oral PARP1-selective inhibitor in this population. The events that occurred most often, and those that most often reached grade 3 or higher, are those for which the protocol required specific monitoring and management. No new safety concern was identified.

13 DISCUSSION AND OVERALL CONCLUSIONS

Study KLV-7701 evaluated kelvaparib in recurrent ovarian carcinoma after prior platinum-based therapy, platinum-sensitive or platinum-resistant. At this data cut-off, before the final analysis, progression-free survival favoured kelvaparib over placebo (stratified hazard ratio 0.72, 95 per cent CI 0.55 to 0.94; stratified log-rank p = 0.0140); the result is descriptive, because the statistical analysis plan spends no alpha before the final analysis. The overall response rate of 50.4 per cent is also descriptive. The safety profile was dominated by anaemia and myelosuppression, both on-target consequences of the mechanism of action. Myelodysplastic syndrome or acute myeloid leukaemia occurred in 1.4 per cent of subjects (Section 12.2.6). Limitations should be borne in mind. The study was open-label, which may influence investigator assessment of response and reporting of subjective adverse events. Follow-up was ongoing at the data cut-off and time-to-event estimates are therefore immature. Taken together, the results of study KLV-7701 support the continued development of kelvaparib at 300 mg, with weekly blood counts for the first month and monthly thereafter, in high-grade serous ovarian carcinoma.

The population enrolled was consistent with the intended population and with the eligibility criteria. Exposure was sufficient to characterise both the efficacy and the safety of the regimen. The primary analysis was performed as planned; the sensitivity analyses the statistical analysis plan lists are not reported here.

The population enrolled was the population the protocol described. Median age, performance status and prior therapy were consistent with the eligibility criteria, and there is no indication that the study enrolled a population easier to treat than the one intended.

Exposure was adequate. The median relative dose intensity indicates that the regimen as written could be delivered, and the pattern of reduction and interruption was consistent with the toxicities anticipated for oral PARP1-selective inhibitor.

The efficacy result should be read together with its confidence interval and with the maturity of the follow-up. Because the study was randomised, the comparison is internally controlled and the estimate does not depend on an external assumption about how this population would otherwise have fared.

The safety profile was consistent with the mechanism and with the earlier clinical experience. The events reported most often, and those most often reaching grade 3 or higher, were those for which the protocol required specific monitoring, which indicates that the monitoring was directed

at the right things.

84 subjects (14.2 per cent) discontinued study treatment because of an adverse event. That figure, rather than the overall incidence of events, is the measure of whether the regimen is tolerable in practice, because it counts the events that could not be managed by dose modification.

Deaths were reviewed individually. Those attributed to an adverse event are described in the narratives; the remainder are recorded as disease outcomes and are consistent with what would be expected in this population over this period.

The limitations of this study should be borne in mind. Follow-up at this analysis is limited and the longer time-to-event estimates are immature. The study was open-label, so investigator assessment of any subjective endpoint may have been influenced by knowledge of the assigned treatment. The subgroup analyses are descriptive.

Taken together, the results support the conclusions stated above and do not identify a safety concern that was not anticipated. Nothing in this analysis indicates that the assumptions underlying the design were materially wrong.

The limitations of the study should be borne in mind when interpreting these results. The study was open-label, so investigator assessment of subjective endpoints may have been influenced by knowledge of the assigned treatment. Follow-up at the time of this analysis is limited, so the estimates of the longer time-to-event endpoints are immature and will change as further events accrue.

14 TABLES AND FIGURES REFERRED TO BUT NOT IN THE TEXT

The summary tables referred to in the text are presented in this section. Subject data listings are provided in Appendix 16.2.

14.1 Demographic data

Table 14.1.1 Demographic and Baseline Characteristics – Safety Analysis Set

Characteristic	Kelvaparib 300 mg (N=394)	Placebo (N=197)	Total (N=591)
Age (years)			
Mean (SD)	65.3 (9.00)	66.2 (7.96)	65.4 (8.84)
Median	65.0	66.0	65.5
Min, max	38, 88	45, 88	38, 88
>= 65 years, n (%)	225 (53.8)	50 (59.5)	275 (54.8)
Sex, n (%)			
Male	0 (0.0)	0 (0.0)	0 (0.0)
Female	394 (100.0)	197 (100.0)	591 (100.0)
Race, n (%)			
White	342 (81.8)	63 (75.0)	405 (80.7)
Asian	38 (9.1)	10 (11.9)	48 (9.6)
Black or African American	20 (4.8)	5 (6.0)	25 (5.0)
Other	18 (4.3)	6 (7.1)	24 (4.8)
ECOG performance status, n (%)			
0	193 (46.2)	42 (50.0)	235 (46.8)
1	184 (44.0)	34 (40.5)	218 (43.4)
2	41 (9.8)	8 (9.5)	49 (9.8)
BRCA mutation status, n (%)			
BRCA1 mutated	101 (24.2)	21 (25.0)	122 (24.3)
BRCA2 mutated	62 (14.8)	10 (11.9)	72 (14.3)
BRCA wild-type	255 (61.0)	53 (63.1)	308 (61.4)
Homologous recombination deficiency, n (%)	222 (53.1)	41 (48.8)	263 (52.4)
Platinum-sensitive disease, n (%)	265 (63.4)	52 (61.9)	317 (63.1)
Residual disease after cytoreductive surgery, n (%)	178 (42.6)	36 (42.9)	214 (42.6)
Number of prior lines of platinum-based therapy			
Median	2.0	2.0	2.0
>= 3, n (%)	122 (29.2)	19 (22.6)	141 (28.1)

SD = standard deviation; ECOG = Eastern Cooperative Oncology Group. Percentages are based on the number of subjects in the safety analysis set within each group.

Table 14.1.2 Subject Disposition – Safety Analysis Set

Disposition	Kelvaparib 300 mg (N=394)	Placebo (N=197)	Total (N=591)
Subjects treated, n	394	197	591
Treatment ongoing at data cut-off, n (%)	89 (22.6)	34 (17.3)	123 (20.8)
Discontinued study treatment, n (%)	317 (75.8)	67 (79.8)	384 (76.5)
Disease progression	30 (7.6)	35 (17.8)	65 (11.0)
Adverse event	25 (6.3)	11 (5.6)	36 (6.1)
Death	31 (7.9)	19 (9.6)	50 (8.5)
Withdrawal by subject	61 (15.5)	27 (13.7)	88 (14.9)
Physician decision	50 (12.7)	24 (12.2)	74 (12.5)
Lost to follow-up	30 (7.6)	8 (4.1)	38 (6.4)
Protocol deviation	24 (6.1)	10 (5.1)	34 (5.8)

Percentages are based on the number of subjects treated within each group.

Table 14.1.3 Extent of Exposure – Safety Analysis Set

Exposure parameter	Kelvaparib 300 mg (N=394)	Placebo (N=197)	Total (N=591)
Duration of treatment (days)			
Median	112.0	114.0	114.0
Q1, Q3	63.2, 159.0	55.0, 168.0	60.0, 163.5
Min, max	8, 340	3, 353	3, 353
Cycles started, median	4.0	4.0	4.0
Relative dose intensity (%), median	86.8	88.1	87.0
Any dose reduction, n (%)	69 (17.5)	37 (18.8)	106 (17.9)
Any dose interruption, n (%)	110 (27.9)	52 (26.4)	162 (27.4)

Q1 = first quartile; Q3 = third quartile. Relative dose intensity is the total dose received divided by the total dose planned over the treatment period.

14.2 Efficacy data

Table 14.2.1 Best Overall Response by RECIST 1.1 – Full Analysis Set

Best overall response	Kelvaparib 300 mg (N=381)	Placebo (N=188)	Total (N=569)
Complete response (CR)	40 (10.5)	14 (7.4)	54 (9.5)
Partial response (PR)	178 (46.7)	55 (29.3)	233 (40.9)
Stable disease	91 (23.9)	64 (34.0)	155 (27.2)
Progressive disease	36 (9.4)	37 (19.7)	73 (12.8)
Not evaluable	36 (9.4)	18 (9.6)	54 (9.5)
Objective response rate, n (%)	218 (57.2)	69 (36.7)	287 (50.4)
95% CI	(52.1, 62.2)	(29.8, 44.0)	(46.3, 54.6)

CI = confidence interval. Assessed according to RECIST version 1.1. Confidence intervals are exact (Clopper-Pearson).

Table 14.2.2 Kaplan-Meier Estimates of Time-to-event Endpoints – Full Analysis Set

Endpoint	Kelvaparib 300 mg (N=381)	Placebo (N=188)	Total (N=569)
Duration of response, median (months)	12.0	9.3	12.0
25th, 75th percentile	(5.5, 21.8)	(2.9, 13.7)	(5.3, 15.9)
Progression-free survival, median (months)	9.0	6.8	8.2
25th, 75th percentile	(3.6, 14.8)	(2.8, 13.9)	(3.3, 14.5)
Overall survival, median (months)	27.2	NE	27.2
25th, 75th percentile	(22.5, NE)	NE	(22.5, NE)

NE = not estimable. Duration of response is analysed among responding subjects only. It is measured from the first dose to the end of response recorded in the analysis dataset; response and progression dates cannot be re-derived because no assessment-level tumour data were collected in the database.

Table 14.2.3 Overall Response Rate and Progression-free Survival by BRCA Mutation Status – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
BRCA wild-type	347	177 (51.0)	(45.6, 56.4)	8.2
BRCA1 mutated	136	67 (49.3)	(40.6, 58.0)	7.6
BRCA2 mutated	86	43 (50.0)	(39.0, 61.0)	10.3

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 14.2.4 Overall Response Rate and Progression-free Survival by Homologous Recombination Status – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
Negative	277	132 (47.7)	(41.6, 53.7)	7.6
Positive	292	155 (53.1)	(47.2, 58.9)	8.7

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 14.2.5 Overall Response Rate and Progression-free Survival by Platinum Sensitivity – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
No	204	110 (53.9)	(46.8, 60.9)	8.4
Yes	365	177 (48.5)	(43.3, 53.8)	8.1

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 14.2.6 Overall Response Rate and Progression-free Survival by Residual Disease After Surgery – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
Absent	331	167 (50.5)	(44.9, 56.0)	8.5
Present	238	120 (50.4)	(43.9, 56.9)	7.5

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 14.2.7 Overall Response Rate and Progression-free Survival by Prior Lines of Platinum-based Therapy – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
1 to 2	397	196 (49.4)	(44.3, 54.4)	8.5
>= 3	172	91 (52.9)	(45.2, 60.5)	7.1

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 14.2.8 Overall Response Rate and Progression-free Survival by Age Category – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
< 65 years	260	143 (55.0)	(48.7, 61.2)	8.1
>= 65 years	309	144 (46.6)	(40.9, 52.3)	8.4

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 14.2.9 Overall Response Rate and Progression-free Survival by ECOG Performance Status – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
0	266	139 (52.3)	(46.1, 58.4)	7.6
1	252	122 (48.4)	(42.1, 54.8)	9.1
2	51	26 (51.0)	(36.6, 65.2)	8.4

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

14.3 Safety data

Table 14.3.1 Overview of Treatment-emergent Adverse Events – Safety Analysis Set

Category, n (%)	Kelvaparib 300 mg (N=394)	Placebo (N=197)	Total (N=591)
Any TEAE	391 (99.2)	197 (100.0)	588 (99.5)
Any treatment-related TEAE	383 (97.2)	190 (96.4)	573 (97.0)
Any grade >= 3 TEAE	172 (43.7)	97 (49.2)	269 (45.5)
Any serious TEAE	86 (21.8)	52 (26.4)	138 (23.4)
Any TEAE leading to dose reduction	69 (17.5)	36 (18.3)	105 (17.8)
Any TEAE leading to discontinuation	54 (13.7)	30 (15.2)	84 (14.2)
Any fatal TEAE	36 (9.1)	21 (10.7)	57 (9.6)

TEAE = treatment-emergent adverse event. A subject is counted once within each category.

Table 14.3.2 Treatment-emergent Adverse Events Occurring in at Least 2% of Subjects by System Organ Class and Preferred Term – Safety Analysis Set

System organ class / Preferred term	Kelvaparib 300 mg (N=394)	Placebo (N=197)	Total (N=591)
Gastrointestinal disorders	335 (85.0)	166 (84.3)	501 (84.8)
Nausea	236 (59.9)	123 (62.4)	359 (60.7)
Vomiting	103 (26.1)	47 (23.9)	150 (25.4)
Constipation	94 (23.9)	55 (27.9)	149 (25.2)
Diarrhoea	74 (18.8)	34 (17.3)	108 (18.3)
Abdominal pain	75 (19.0)	32 (16.2)	107 (18.1)
Dyspepsia	39 (9.9)	19 (9.6)	58 (9.8)
Blood and lymphatic system disorders	288 (73.1)	145 (73.6)	433 (73.3)
Anaemia	185 (47.0)	99 (50.3)	284 (48.1)
Thrombocytopenia	97 (24.6)	47 (23.9)	144 (24.4)
Neutropenia	75 (19.0)	40 (20.3)	115 (19.5)
Leukopenia	43 (10.9)	26 (13.2)	69 (11.7)
Lymphopenia	23 (5.8)	19 (9.6)	42 (7.1)
General disorders and administration site conditions	236 (59.9)	118 (59.9)	354 (59.9)
Fatigue	185 (47.0)	90 (45.7)	275 (46.5)
Asthenia	50 (12.7)	31 (15.7)	81 (13.7)
Oedema peripheral	29 (7.4)	16 (8.1)	45 (7.6)
Pyrexia	20 (5.1)	8 (4.1)	28 (4.7)
Nervous system disorders	169 (42.9)	84 (42.6)	253 (42.8)
Headache	74 (18.8)	38 (19.3)	112 (19.0)
Dysgeusia	61 (15.5)	42 (21.3)	103 (17.4)
Dizziness	67 (17.0)	26 (13.2)	93 (15.7)
Musculoskeletal and connective tissue disorders	107 (27.2)	61 (31.0)	168 (28.4)
Back pain	45 (11.4)	26 (13.2)	71 (12.0)
Arthralgia	43 (10.9)	22 (11.2)	65 (11.0)
Myalgia	25 (6.3)	16 (8.1)	41 (6.9)
Infections and infestations	108 (27.4)	58 (29.4)	166 (28.1)
Nasopharyngitis	51 (12.9)	21 (10.7)	72 (12.2)
Urinary tract infection	35 (8.9)	23 (11.7)	58 (9.8)
Pneumonia	24 (6.1)	13 (6.6)	37 (6.3)
Sepsis	9 (2.3)	10 (5.1)	19 (3.2)
Investigations	100 (25.4)	53 (26.9)	153 (25.9)
Alanine aminotransferase increased	34 (8.6)	18 (9.1)	52 (8.8)
Aspartate aminotransferase increased	26 (6.6)	15 (7.6)	41 (6.9)

System organ class / Preferred term	Kelvaparib 300 mg (N=394)	Placebo (N=197)	Total (N=591)
Blood creatinine increased	25 (6.3)	14 (7.1)	39 (6.6)
Weight decreased	27 (6.9)	12 (6.1)	39 (6.6)
Metabolism and nutrition disorders	90 (22.8)	54 (27.4)	144 (24.4)
Decreased appetite	90 (22.8)	54 (27.4)	144 (24.4)
Respiratory, thoracic and mediastinal disorders	91 (23.1)	46 (23.4)	137 (23.2)
Cough	52 (13.2)	25 (12.7)	77 (13.0)
Dyspnoea	33 (8.4)	16 (8.1)	49 (8.3)
Pneumonitis	13 (3.3)	7 (3.6)	20 (3.4)
Psychiatric disorders	41 (10.4)	27 (13.7)	68 (11.5)
Insomnia	41 (10.4)	27 (13.7)	68 (11.5)
Skin and subcutaneous tissue disorders	30 (7.6)	25 (12.7)	55 (9.3)
Rash	30 (7.6)	25 (12.7)	55 (9.3)
Vascular disorders	25 (6.3)	9 (4.6)	34 (5.8)
Hypertension	25 (6.3)	9 (4.6)	34 (5.8)

Subjects are counted once per preferred term and once per system organ class at their highest reported grade. MedDRA version 25.0.

Table 14.3.3 Treatment-emergent Adverse Events of Grade 3 or Higher by System Organ Class and Preferred Term – Safety Analysis Set

System organ class / Preferred term	Kelvaparib 300 mg (N=394)	Placebo (N=197)	Total (N=591)
Blood and lymphatic system disorders	100 (25.4)	56 (28.4)	156 (26.4)
Anaemia	53 (13.5)	37 (18.8)	90 (15.2)
Thrombocytopenia	21 (5.3)	11 (5.6)	32 (5.4)
Neutropenia	21 (5.3)	9 (4.6)	30 (5.1)
Lymphopenia	6 (1.5)	2 (1.0)	8 (1.4)
Leukopenia	4 (1.0)	4 (2.0)	8 (1.4)
Gastrointestinal disorders	25 (6.3)	14 (7.1)	39 (6.6)
Nausea	11 (2.8)	7 (3.6)	18 (3.0)
Vomiting	6 (1.5)	2 (1.0)	8 (1.4)
Intestinal obstruction	2 (0.5)	2 (1.0)	4 (0.7)
Ascites	2 (0.5)	1 (0.5)	3 (0.5)
Abdominal pain	2 (0.5)	1 (0.5)	3 (0.5)
Constipation	1 (0.3)	1 (0.5)	2 (0.3)
Diarrhoea	2 (0.5)	0 (0.0)	2 (0.3)
Infections and infestations	24 (6.1)	14 (7.1)	38 (6.4)
Pneumonia	13 (3.3)	4 (2.0)	17 (2.9)
Sepsis	8 (2.0)	8 (4.1)	16 (2.7)
Urinary tract infection	2 (0.5)	2 (1.0)	4 (0.7)
Nasopharyngitis	1 (0.3)	0 (0.0)	1 (0.2)
General disorders and administration site conditions	14 (3.6)	8 (4.1)	22 (3.7)
Fatigue	8 (2.0)	7 (3.6)	15 (2.5)
Asthenia	4 (1.0)	0 (0.0)	4 (0.7)
Pyrexia	2 (0.5)	0 (0.0)	2 (0.3)
Oedema peripheral	0 (0.0)	1 (0.5)	1 (0.2)
Neoplasms benign, malignant and unspecified	8 (2.0)	8 (4.1)	16 (2.7)
Malignant neoplasm progression	5 (1.3)	5 (2.5)	10 (1.7)
Myelodysplastic syndrome	3 (0.8)	3 (1.5)	6 (1.0)
Respiratory, thoracic and mediastinal disorders	8 (2.0)	8 (4.1)	16 (2.7)
Pneumonitis	3 (0.8)	3 (1.5)	6 (1.0)
Dyspnoea	4 (1.0)	2 (1.0)	6 (1.0)
Pulmonary embolism	1 (0.3)	1 (0.5)	2 (0.3)
Pleural effusion	0 (0.0)	1 (0.5)	1 (0.2)
Cough	0 (0.0)	1 (0.5)	1 (0.2)

System organ class / Preferred term	Kelvaparib 300 mg (N=394)	Placebo (N=197)	Total (N=591)
Investigations	8 (2.0)	4 (2.0)	12 (2.0)
Alanine aminotransferase increased	2 (0.5)	4 (2.0)	6 (1.0)
Blood creatinine increased	5 (1.3)	0 (0.0)	5 (0.8)
Aspartate aminotransferase increased	2 (0.5)	0 (0.0)	2 (0.3)
Nervous system disorders	2 (0.5)	3 (1.5)	5 (0.8)
Headache	2 (0.5)	1 (0.5)	3 (0.5)
Dizziness	0 (0.0)	2 (1.0)	2 (0.3)
Vascular disorders	3 (0.8)	0 (0.0)	3 (0.5)
Hypertension	3 (0.8)	0 (0.0)	3 (0.5)
Metabolism and nutrition disorders	1 (0.3)	2 (1.0)	3 (0.5)
Decreased appetite	1 (0.3)	2 (1.0)	3 (0.5)
Skin and subcutaneous tissue disorders	2 (0.5)	0 (0.0)	2 (0.3)
Rash	2 (0.5)	0 (0.0)	2 (0.3)
Musculoskeletal and connective tissue disorders	1 (0.3)	0 (0.0)	1 (0.2)
Arthralgia	1 (0.3)	0 (0.0)	1 (0.2)
Psychiatric disorders	1 (0.3)	0 (0.0)	1 (0.2)
Insomnia	1 (0.3)	0 (0.0)	1 (0.2)

Subjects are counted once per preferred term and once per system organ class at their highest reported grade. MedDRA version 25.0.

Table 14.3.4 Treatment-related Treatment-emergent Adverse Events Occurring in at Least 2% of Subjects – Safety Analysis Set

Preferred term	Kelvaparib 300 mg (N=394)	Placebo (N=197)	Total (N=591)
Nausea	165 (41.9)	77 (39.1)	242 (40.9)
Anaemia	105 (26.6)	64 (32.5)	169 (28.6)
Fatigue	99 (25.1)	46 (23.4)	145 (24.5)
Vomiting	67 (17.0)	31 (15.7)	98 (16.6)
Constipation	63 (16.0)	31 (15.7)	94 (15.9)
Decreased appetite	60 (15.2)	31 (15.7)	91 (15.4)
Thrombocytopenia	54 (13.7)	28 (14.2)	82 (13.9)
Headache	50 (12.7)	28 (14.2)	78 (13.2)
Neutropenia	47 (11.9)	28 (14.2)	75 (12.7)
Diarrhoea	45 (11.4)	26 (13.2)	71 (12.0)
Abdominal pain	42 (10.7)	23 (11.7)	65 (11.0)
Dysgeusia	38 (9.6)	23 (11.7)	61 (10.3)
Dizziness	39 (9.9)	20 (10.2)	59 (10.0)
Asthenia	34 (8.6)	18 (9.1)	52 (8.8)
Cough	34 (8.6)	17 (8.6)	51 (8.6)
Nasopharyngitis	33 (8.4)	14 (7.1)	47 (8.0)
Back pain	28 (7.1)	18 (9.1)	46 (7.8)
Insomnia	27 (6.9)	14 (7.1)	41 (6.9)
Leukopenia	25 (6.3)	16 (8.1)	41 (6.9)
Urinary tract infection	26 (6.6)	13 (6.6)	39 (6.6)
Rash	18 (4.6)	20 (10.2)	38 (6.4)
Dyspepsia	24 (6.1)	13 (6.6)	37 (6.3)
Arthralgia	21 (5.3)	16 (8.1)	37 (6.3)
Alanine aminotransferase increased	19 (4.8)	12 (6.1)	31 (5.2)
Dyspnoea	23 (5.8)	6 (3.0)	29 (4.9)
Blood creatinine increased	18 (4.6)	10 (5.1)	28 (4.7)
Oedema peripheral	20 (5.1)	6 (3.0)	26 (4.4)
Weight decreased	19 (4.8)	7 (3.6)	26 (4.4)
Hypertension	18 (4.6)	7 (3.6)	25 (4.2)
Myalgia	16 (4.1)	9 (4.6)	25 (4.2)
Lymphopenia	10 (2.5)	13 (6.6)	23 (3.9)
Aspartate aminotransferase increased	14 (3.6)	9 (4.6)	23 (3.9)
Pneumonia	15 (3.8)	8 (4.1)	23 (3.9)
Sepsis	7 (1.8)	10 (5.1)	17 (2.9)
Pyrexia	11 (2.8)	3 (1.5)	14 (2.4)

Relationship assessed by the investigator. Percentages are based on the number of subjects treated in each group. Overall N = 591.

Table 14.3.5 Serious Treatment-emergent Adverse Events by Preferred Term – Safety Analysis Set

Preferred term	Kelvaparib 300 mg (N=394)	Placebo (N=197)	Total (N=591)
Anaemia	31 (7.9)	21 (10.7)	52 (8.8)
Pneumonia	11 (2.8)	3 (1.5)	14 (2.4)
Thrombocytopenia	12 (3.0)	1 (0.5)	13 (2.2)
Malignant neoplasm progression	5 (1.3)	5 (2.5)	10 (1.7)
Sepsis	5 (1.3)	4 (2.0)	9 (1.5)
Neutropenia	4 (1.0)	3 (1.5)	7 (1.2)
Pneumonitis	3 (0.8)	3 (1.5)	6 (1.0)
Myelodysplastic syndrome	3 (0.8)	2 (1.0)	5 (0.8)
Fatigue	3 (0.8)	1 (0.5)	4 (0.7)
Intestinal obstruction	2 (0.5)	2 (1.0)	4 (0.7)
Ascites	2 (0.5)	1 (0.5)	3 (0.5)
Nausea	2 (0.5)	1 (0.5)	3 (0.5)
Alanine aminotransferase increased	1 (0.3)	1 (0.5)	2 (0.3)
Dyspnoea	2 (0.5)	0 (0.0)	2 (0.3)
Leukopenia	0 (0.0)	2 (1.0)	2 (0.3)
Pulmonary embolism	1 (0.3)	1 (0.5)	2 (0.3)
Aspartate aminotransferase increased	1 (0.3)	0 (0.0)	1 (0.2)
Decreased appetite	0 (0.0)	1 (0.5)	1 (0.2)
Diarrhoea	1 (0.3)	0 (0.0)	1 (0.2)
Dizziness	0 (0.0)	1 (0.5)	1 (0.2)
Hypertension	1 (0.3)	0 (0.0)	1 (0.2)
Lymphopenia	1 (0.3)	0 (0.0)	1 (0.2)
Pleural effusion	0 (0.0)	1 (0.5)	1 (0.2)
Urinary tract infection	1 (0.3)	0 (0.0)	1 (0.2)
Vomiting	1 (0.3)	0 (0.0)	1 (0.2)

Subjects are counted once per preferred term. Percentages are based on N = 591.

Table 14.3.6 Treatment-emergent Adverse Events Leading to Discontinuation of Study Treatment – Safety Analysis Set

Preferred term	Kelvaparib 300 mg (N=394)	Placebo (N=197)	Total (N=591)
Anaemia	9 (2.3)	6 (3.0)	15 (2.5)
Malignant neoplasm progression	5 (1.3)	5 (2.5)	10 (1.7)
Sepsis	6 (1.5)	3 (1.5)	9 (1.5)
Pneumonia	7 (1.8)	1 (0.5)	8 (1.4)
Neutropenia	5 (1.3)	2 (1.0)	7 (1.2)
Thrombocytopenia	6 (1.5)	0 (0.0)	6 (1.0)
Intestinal obstruction	2 (0.5)	2 (1.0)	4 (0.7)
Ascites	2 (0.5)	1 (0.5)	3 (0.5)
Dyspnoea	3 (0.8)	0 (0.0)	3 (0.5)
Leukopenia	1 (0.3)	2 (1.0)	3 (0.5)
Myelodysplastic syndrome	1 (0.3)	2 (1.0)	3 (0.5)
Fatigue	2 (0.5)	0 (0.0)	2 (0.3)
Lymphopenia	2 (0.5)	0 (0.0)	2 (0.3)
Pneumonitis	0 (0.0)	2 (1.0)	2 (0.3)
Pulmonary embolism	1 (0.3)	1 (0.5)	2 (0.3)
Abdominal pain	1 (0.3)	0 (0.0)	1 (0.2)
Asthenia	1 (0.3)	0 (0.0)	1 (0.2)
Constipation	0 (0.0)	1 (0.5)	1 (0.2)
Decreased appetite	1 (0.3)	0 (0.0)	1 (0.2)
Dizziness	0 (0.0)	1 (0.5)	1 (0.2)
Nausea	1 (0.3)	0 (0.0)	1 (0.2)
Pleural effusion	0 (0.0)	1 (0.5)	1 (0.2)
Pyrexia	1 (0.3)	0 (0.0)	1 (0.2)

Percentages are based on N = 591.

Table 14.3.7 Deaths – Safety Analysis Set

Category, n (%)	Kelvaparib 300 mg (N=394)	Placebo (N=197)	Total (N=591)
Any death	61 (15.5)	31 (15.7)	92 (15.6)
Death within 30 days of last dose	31 (7.9)	19 (9.6)	50 (8.5)
Primary cause of death			
Disease progression	23 (5.8)	8 (4.1)	31 (5.2)
Adverse event	36 (9.1)	21 (10.7)	57 (9.6)
Other	2 (0.5)	2 (1.0)	4 (0.7)

Percentages are based on the number of subjects treated within each group.

Table 14.3.8 Shift from Baseline to Worst Post-baseline CTCAE Grade for Selected Laboratory Parameters – Safety Analysis Set

Laboratory parameter	Kelvaparib 300 mg: worst grade 3 n (%)	Kelvaparib 300 mg: worst grade 4 n (%)	Kelvaparib 300 mg: shift >= 2 grades n (%)	Placebo: worst grade 3 n (%)	Placebo: worst grade 4 n (%)	Placebo: shift >= 2 grades n (%)	Total: worst grade 3 n (%)	Total: worst grade 4 n (%)	Total: shift >= 2 grades n (%)
Haemoglobin	60 (14.4)	15 (3.6)	137 (32.8)	14 (16.7)	6 (7.1)	38 (45.2)	74 (14.7)	21 (4.2)	175 (34.9)
Platelets	30 (7.2)	7 (1.7)	80 (19.1)	6 (7.1)	1 (1.2)	12 (14.3)	36 (7.2)	8 (1.6)	92 (18.3)
Neutrophils	35 (8.4)	16 (3.8)	84 (20.1)	8 (9.5)	4 (4.8)	17 (20.2)	43 (8.6)	20 (4.0)	101 (20.1)
Lymphocytes	34 (8.1)	12 (2.9)	74 (17.7)	5 (6.0)	1 (1.2)	15 (17.9)	39 (7.8)	13 (2.6)	89 (17.7)
CA-125	55 (13.2)	12 (2.9)	136 (32.5)	19 (22.6)	4 (4.8)	33 (39.3)	74 (14.7)	16 (3.2)	169 (33.7)
ALT	23 (5.5)	3 (0.7)	50 (12.0)	6 (7.1)	0 (0.0)	10 (11.9)	29 (5.8)	3 (0.6)	60 (12.0)
AST	14 (3.3)	1 (0.2)	33 (7.9)	2 (2.4)	1 (1.2)	7 (8.3)	16 (3.2)	2 (0.4)	40 (8.0)
Creatinine	22 (5.3)	8 (1.9)	49 (11.7)	4 (4.8)	0 (0.0)	4 (4.8)	26 (5.2)	8 (1.6)	53 (10.6)
Lactate dehydrogenase	26 (6.2)	5 (1.2)	58 (13.9)	3 (3.6)	2 (2.4)	6 (7.1)	29 (5.8)	7 (1.4)	64 (12.7)
Albumin	14 (3.3)	4 (1.0)	27 (6.5)	4 (4.8)	1 (1.2)	9 (10.7)	18 (3.6)	5 (1.0)	36 (7.2)

CTCAE = Common Terminology Criteria for Adverse Events, version 5.0. Percentages are based on the number of subjects treated in each group. Overall N = 591.

Table 14.3.9 Time to Onset and Resolution of the Most Frequent Treatment-emergent Adverse Events – Safety Analysis Set

Preferred term	Events, n	Median time to onset (days)	Resolved, n (%)	Median time to resolution (days)
Nausea	359	31.0	297 (82.7)	11.0
Anaemia	285	27.0	238 (83.5)	10.0
Fatigue	275	32.0	222 (80.7)	10.0
Vomiting	150	32.0	117 (78.0)	11.0
Constipation	149	28.0	122 (81.9)	9.0
Thrombocytopenia	145	40.0	123 (84.8)	12.0
Decreased appetite	144	36.0	119 (82.6)	10.0
Neutropenia	116	35.0	89 (76.7)	10.0
Headache	112	23.5	92 (82.1)	10.0
Diarrhoea	108	37.5	87 (80.6)	11.0
Abdominal pain	107	24.0	92 (86.0)	12.0
Dysgeusia	103	24.0	83 (80.6)	10.0
Dizziness	93	24.0	70 (75.3)	12.5
Asthenia	81	33.0	67 (82.7)	11.0
Cough	77	33.0	61 (79.2)	10.0
Nasopharyngitis	72	27.5	58 (80.6)	11.5
Back pain	71	24.0	60 (84.5)	9.5
Leukopenia	69	29.0	54 (78.3)	11.0
Insomnia	68	37.5	57 (83.8)	8.0
Arthralgia	65	34.0	55 (84.6)	8.0

Events are counted at the event level rather than the subject level. NE = not estimable.

14.3.10 Adverse events within subgroups

The summaries in Tables 14.3.10 to 14.3.16 are exploratory. The subgroups were not chosen or sized for comparisons of safety, the tables are not adjusted for multiplicity, and no conclusion about a difference between subgroups is drawn from them.

Table 14.3.10 Overview of adverse events by BRCA mutation status

Category	BRCA wild-type	BRCA1 mutated	BRCA2 mutated
Any event	359 (99.4)	139 (99.3)	90 (100.0)
Grade 3 or higher	170 (47.1)	62 (44.3)	37 (41.1)
Serious	88 (24.4)	30 (21.4)	20 (22.2)
Led to discontinuation	50 (13.9)	22 (15.7)	12 (13.3)
Fatal	34 (9.4)	16 (11.4)	7 (7.8)

Percentages are calculated on the number of subjects in that subgroup.

Table 14.3.11 Overview of adverse events by homologous recombination status

Category	Negative	Positive
Any event	287 (100.0)	301 (99.0)
Grade 3 or higher	134 (46.7)	135 (44.4)
Serious	72 (25.1)	66 (21.7)
Led to discontinuation	39 (13.6)	45 (14.8)
Fatal	27 (9.4)	30 (9.9)

Percentages are calculated on the number of subjects in that subgroup.

Table 14.3.12 Overview of adverse events by platinum sensitivity

Category	No	Yes
Any event	209 (99.5)	379 (99.5)
Grade 3 or higher	92 (43.8)	177 (46.5)
Serious	52 (24.8)	86 (22.6)
Led to discontinuation	28 (13.3)	56 (14.7)
Fatal	18 (8.6)	39 (10.2)

Percentages are calculated on the number of subjects in that subgroup.

Table 14.3.13 Overview of adverse events by residual disease after surgery

Category	Absent	Present
Any event	338 (99.4)	250 (99.6)
Grade 3 or higher	151 (44.4)	118 (47.0)
Serious	72 (21.2)	66 (26.3)
Led to discontinuation	41 (12.1)	43 (17.1)
Fatal	27 (7.9)	30 (12.0)

Percentages are calculated on the number of subjects in that subgroup.

Table 14.3.14 Overview of adverse events by prior lines of platinum-based therapy

Category	1 to 2	>= 3
Any event	412 (99.5)	176 (99.4)
Grade 3 or higher	180 (43.5)	89 (50.3)
Serious	85 (20.5)	53 (29.9)
Led to discontinuation	52 (12.6)	32 (18.1)
Fatal	36 (8.7)	21 (11.9)

Percentages are calculated on the number of subjects in that subgroup.

Table 14.3.15 Overview of adverse events by age category

Category	<65	>=65
Any event	272 (100.0)	316 (99.1)
Grade 3 or higher	114 (41.9)	155 (48.6)
Serious	62 (22.8)	76 (23.8)
Led to discontinuation	38 (14.0)	46 (14.4)
Fatal	29 (10.7)	28 (8.8)

Percentages are calculated on the number of subjects in that subgroup.

Table 14.3.16 Overview of adverse events by ECOG performance status

Category	0	1	2
Any event	277 (99.3)	258 (99.6)	53 (100.0)
Grade 3 or higher	132 (47.3)	119 (45.9)	18 (34.0)
Serious	74 (26.5)	57 (22.0)	7 (13.2)
Led to discontinuation	41 (14.7)	38 (14.7)	5 (9.4)
Fatal	29 (10.4)	25 (9.7)	3 (5.7)

Percentages are calculated on the number of subjects in that subgroup.

14.4 Pharmacokinetic data

Table 14.4.1 Summary of Steady-state Pharmacokinetic Parameters – Pharmacokinetic Analysis Set

Treatment	n	Cmax (ng/mL) Geometric mean (%CV)	tmax (h) Median (min, max)	AUC0-12 (ng·h/mL) Geometric mean (%CV)	t1/2 (h) Mean (SD)	CL/F (L/h) Geometric mean (%CV)
Kelvaparib 300 mg	394	1770 (29.4)	2.0 (2.0, 2.0)	17709 (35.8)	14.4 (6.63)	16.94 (35.8)

AUC0-12 = area under the concentration-time curve over one dosing interval; CL/F = apparent oral clearance; Cmax = maximum observed concentration; %CV = geometric coefficient of variation; t1/2 = terminal elimination half-life; tmax = time to maximum concentration.

Table 14.4.2 Accumulation Ratio and Trough Concentration – Pharmacokinetic Analysis Set

Treatment	n	Accumulation ratio Geometric mean (%CV)	Ctrough (ng/mL) Geometric mean (%CV)
Kelvaparib 300 mg	394	2.84 (25.2)	1087.4 (49.1)

Accumulation ratio is the ratio of AUC0-24 at steady state to AUC0-24 after the first dose.

15 REFERENCE LIST

- Eisenhauer EA, et al. New response evaluation criteria in solid tumours: revised RECIST guideline version 1.1. Eur J Cancer. 2009;45(2):228-247.
- Rustin GJS, et al. Definitions for response and progression in ovarian cancer using CA-125: GCIG consensus. Int J Gynecol Cancer. 2011;21(2):419-423.
- Common Terminology Criteria for Adverse Events version 5.0. National Cancer Institute; 2017.
- International Council for Harmonisation. Structure and content of clinical study reports E3. 1995.
- Brookmeyer R, Crowley J. A confidence interval for the median survival time. Biometrics. 1982;38(1):29-41.

16 APPENDICES

16.1.1	Protocol and protocol amendments
16.1.2	Sample case report form
16.1.3	List of independent ethics committees and institutional review boards
16.1.4	List of investigators and study sites
16.1.5	Signatures of principal investigators
16.1.6	Listing of subjects receiving test drug from specific batches
16.1.7	Randomisation scheme and codes
16.1.8	Audit certificates
16.1.9	Documentation of statistical methods
16.1.10	Documentation of inter-laboratory standardisation and quality assurance
16.1.11	Publications based on the study
16.1.12	Important publications referenced in the report
16.2.1	Listing of subjects who discontinued study treatment
16.2.2	Listing of protocol deviations
16.2.3	Listing of subjects excluded from the efficacy analysis
16.2.4	Listing of demographic data
16.2.5	Listing of study treatment administration and compliance
16.2.6	Listing of individual drug concentration data
16.2.7	Listing of individual efficacy response data
16.2.8	Listing of individual time to event data
16.2.9	Listing of adverse events
16.2.10	Listing of individual laboratory measurements
16.2.11	Listing of vital signs
16.2.12	Listing of concomitant medications
16.2.13	Listing of subject disposition
16.3	Case report forms (4 volumes)
16.4	Individual subject data listings (archival): not applicable; individual data are provided as the analysis datasets